

Data Provenance and Evaluation Validity in Alzheimer’s Disease MRI Deep Learning: Controlled Leakage and Provenance-Degradation Experiments (SplitGuard-AD)

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Abstract

Deep-learning models for Alzheimer’s disease (AD) classification from structural MRI routinely report near-perfect benchmark performance that does not survive patient-independent evaluation. We treat this as a data-provenance problem and measure both links of the dependency. On ADNI1 (**220** cognitively normal (CN) and AD participants, **1,123** scans, one 2D coronal-centre slice per scan), a random-split area under the receiver operating characteristic curve (AUROC) of 0.947 falls to 0.829 under component-safe evaluation: an inflation gap of +0.118 [+0.096, +0.145], replicated on OASIS-1 (+0.035) and under a DenseNet-121 backbone. Two controlled interventions calibrate it. Injecting contaminated test scans gives **+0.125** AUROC per unit contamination. Deleting participant identifiers shows that a leakage graph prevents **49%** of the residual leakage subject-wise splitting admits at **10%** identifier loss, where that tenth alone returns **36%** of test participants to training. A third instance is observed rather than simulated: a widely used public 2D AD benchmark is, by content matching, a redistribution of **200** OASIS-1 participants whose identifiers were replaced by a filename index that merges and splits them. Grouping by that index, the release’s only identifier, still leaves **70%** of test images with a participant seen in training

and +0.027 AUROC of inflation. Upstream, **22%** of the ADNI labels rest on a **180**-day date-proximity join rather than an exact visit key. At **0.90** specificity the ADNI gap is 45.7 additional missed diagnoses per **1,000** patients screened. Split manifests, component tables, provenance-recovery scripts and the rebuild pipeline are released under Apache-2.0.

Keywords: Patient-identity leakage, Evaluation protocol, Alzheimer’s disease, Structural MRI, Data provenance, Reproducibility

1 Introduction

Alzheimer’s disease (AD) is the leading cause of dementia worldwide and is a major target for medical-imaging deep learning. The modelling literature is mature: 3D residual networks [1], hierarchical fully-convolutional architectures that jointly localise atrophy and classify [2], and interpretability-aware frameworks validated against neuropathology [3]. Hundreds of studies report near-perfect benchmark accuracy on structural MRI [4–6]. These reports, however, rarely survive patient-independent evaluations. The primary cause is *patient-identity leakage* across train and test partitions, which arises in AD MRI through several distinct channels (visualised in Figure 1): (i) multiple 2D slices from the same 3D volume distributed across splits (within-volume slice leakage); (ii) different visits from the same patient’s longitudinal record distributed across splits (longitudinal same-subject leakage); (iii) near-duplicate or augmentation-derived images placed on opposite sides of the split (near-duplicate leakage); and (iv) rescans or re-acquisitions of the same session split across partitions (session/re-acquisition leakage). Because AD MRI research routinely uses longitudinal cohorts with multiple visits per patient (ADNI: median 5 scans/subject) as well as slice-based 2D representations, each of these channels is a live risk in practice. Under any of them, the trained model recognises images from patients it has already analysed rather than generalising to new patients [6–8].

That leakage inflates reported performance is established. Yagis et al. [8] contrasted slice-wise and subject-wise splitting on four cohorts; Wen et al. [7] reviewed the AD convolutional-network literature and recommended subject-level partitioning; Rosenblatt et al. [9] manipulated five distinct forms of leakage in connectome-based models across four datasets; and a recent scoping review of 44 AD deep-learning studies found a 19-point accuracy gradient between studies judged at low and high risk of leakage [6]. The phenomenon does not need re-demonstrating.

Why patient identity in particular is the channel that matters is itself well established. Shortcut learning [10] occurs when a model exploits stable but clinically irrelevant cues that correlate with the label: hospital-system markers in chest radiographs [11], non-lung cues in COVID classifiers [12], hidden subgroup failures [13]. Structural MRI is unusually exposed to identity-based versions of it. Face-recognition software re-identifies anonymised scans [14], the *BrainPrint* signature is a stable per-subject fingerprint [15], learned representations support cross-cohort subject re-identification [16], and even brain-age prediction carries subject-specific structural

signal [17]. A model evaluated on subject-leaky data can therefore appear to recognise disease while partly recognising the patient. The same protocol-driven reliability gap is documented beyond AD MRI, in COVID imaging [18], in medical-imaging machine learning generally [19], and as a cross-disciplinary leakage taxonomy [20].

Subject-aware splitting is available in general-purpose libraries (scikit-learn `GroupKFold` [21]) and medical-imaging frameworks (MONAI [22]), and a domain-general similarity-aware splitter has been proposed [23]; the CLAIM [24, 25], TRIPOD+AI [26] and PROBAST [27] reporting guidelines all require subject-level splitting for a low risk-of-bias verdict. Protocol B in this paper is operationally `GroupKFold` with `subject_id` as the group key, so every comparison we report against it is a comparison against the practice those guidelines prescribe.

What these studies leave unmeasured is the *transfer function*. Two quantities govern how much optimism a benchmark reports, and neither has been characterised. The first is how much optimism a unit of leakage buys. Rosenblatt et al. come closest, reporting performance deltas at three discrete levels of subject repetition (5, 10 and 20%) in connectome models, but they fit no function across those levels and state no per-unit rate. The second has not been asked at all: what determines how much leakage a given benchmark admits in the first place. That is a question about data provenance. A splitting protocol can only keep a patient’s images together if the identifiers it depends on are present and correct; curated research cohorts supply them, and the redistributed benchmarks the field actually trains on frequently do not.

In this work we measure both quantities on a single cohort, using one experimental template for each: hold everything fixed, sweep one controlled quantity, and read off the curve.

Provenance governs leakage, non-linearly. Deleting subject identifiers from ADNI1 at controlled rates shows how much protection each splitting protocol retains as provenance decays (§3.7). Subject-wise splitting degrades sharply: removing 10% of identifiers returns 36% of test participants to the training partition, a roughly threefold amplification. A leakage graph that also carries session and series identifiers prevents 49% of that residual leakage at 10% loss and 31–37% beyond it, while remaining exactly equivalent to subject-wise splitting when provenance is intact. Both protocols are assigned by the same bin-packer, so the comparison isolates the grouping rather than the splitter. To our knowledge no prior study measures the robustness of a splitting protocol to degraded provenance.

The damage is observable in the wild. A widely used public 2D AD benchmark turns out, by content matching, to be a redistribution of 200 OASIS-1 participants whose identifiers were replaced by a filename index that merges and splits them (§2.3.1). Grouping by that index, the strongest protocol its users can run, still leaves 70% of the test images with a participant seen in training and +0.027 AUROC of inflation standing (§3.1). The deletion experiment above simulates provenance loss; this is a measured instance of it in a benchmark the field trains on.

Leakage buys optimism at a measurable rate. Holding the training set fixed and injecting test-set subject overlap across the full range calibrates the consequence at +0.125 AUROC per unit of contaminated test scans (§3.6). Composing the two curves gives the chain: a tenth of the patient index going missing is worth roughly +0.043

AUROC of optimism under subject-wise splitting and +0.022 under the component-safe protocol, though that composition does not hold at every loss level, for reasons we give in §3.7.

The labels themselves rest on an unaudited join. Of the labels in a standard ADNI1 derivation, 22% are assigned through a 180-day date-proximity match rather than an exact visit key (21% within the CN/AD subset, §2.3.3). The evaluation literature audits how data are split; the metadata join that produces the labels those splits partition has not been audited.

These are anchored to a conventional endpoint measurement. On the ADNI1 CN-versus-AD primary arm (220 participants, 1,123 scans, one 2D coronal-centre slice per scan), random image-level splitting reports an AUROC of 0.947 where component-safe evaluation reports 0.829, an inflation gap of +0.118 (paired-seed interval [+0.096, +0.145], five seeds), replicated directionally on OASIS-1 and under a second backbone (§3.3). Re-expressed at a screening operating point of 0.90 specificity, the same gap corresponds to 45.7 additional missed diagnoses per 1,000 patients screened.

The instrument for all of this is SPLITGUARD-AD, an executable audit layer that builds a per-image leakage graph, emits frozen component-safe split manifests, and gates training on a binary GO/NO-GO decision (§2.2). It is the intervention arm of the provenance experiment rather than a contribution in its own right: on cohorts whose subject identifiers are intact its leakage graph reduces exactly to a subject grouping, with a marginal over GroupKFold-style splitting not distinguishable from zero. It is released with the manifests and component tables behind the splits reported here.

A subject-identity linear probe and a three-point degradation analysis localise the dominant mechanism to biometric leakage: both leaky and honest representations encode patient identity well above chance, and subject-only splitting alone removes most of the gap on every tier (§3.4.1). The framework’s value is therefore not to make models *forget* subject identity, which the representations preserve in either condition, but to deny models the opportunity to *exploit* that identity at test time. Running both protocols on the same cohort additionally surfaces three secondary outputs the leaky benchmark would have hidden, presented as illustrations rather than deployment claims: a cost-of-leakage translation (§3.5.1), a dose-response calibration (§3.6), and an exploratory sex-stratified difference that the leaky protocol flattens (Supplementary Material S6).

The observation that random image-level splits inflate AD MRI scores is not new by itself. Wen *et al.* [7] and Yagis *et al.* [8] quantified the inflation on individual datasets; Rumala [28] showed on longitudinal brain MRI that cross-validation folds which do not separate subjects inflate 3D CNN estimates, and used saliency maps to argue qualitatively that the network attends to subject identity; Ghiasi *et al.* [29] established a leakage-free longitudinal dementia benchmark by withholding the label-defining clinical score and splitting group-aware, reporting the inflated and clean conditions as an on/off pair; general-purpose subject-aware splitting tools such as DataSAIL [23] exist for biomedical data; and biometric-identity work on MRI representations (DeepBrainPrint [16], BrainPrint [15], face-recognition from MRI [14]) has established that structural MRI leaks subject identity at re-identification accuracies far above chance. Across all of it, leakage is treated as a binary hazard: either the

split respects patients or it does not, and the remedy is to group by subject. That framing presumes what it needs, a subject identifier that is present, complete and correct, which is the presumption this paper measures. Table 1 places the present work against that literature on the axes that separate them.

Table 1: What prior leakage studies establish, and what this paper adds. “Dose” is a measured relationship between the amount of contamination and the reported score, rather than a comparison of two endpoints. “Provenance” is whether the identifiers a protocol needs are treated as a variable rather than assumed.

	Inflation shown	Dose measured	Provenance varied	Observed damage	Split manifests
Wen <i>et al.</i> [7] (review)	✓	—	—	—	—
Yagis <i>et al.</i> [8]	✓	—	—	—	—
Rumala [28]	✓	—	—	—	—
Rosenblatt <i>et al.</i> [9]	✓	three levels	—	—	—
Ghiasi <i>et al.</i> [29]	✓	—	—	—	✓
DataSAIL [23] (tool)	—	—	—	—	✓
This paper	✓	continuous	✓	✓	✓

The cross-tier inflation gap is validated on two source cohorts and one redistribution of one of them; the ADNI-specific findings (subject-level aggregation, sex disparity, clinical cost, dose-response) are single-cohort findings on ADNI1 itself, and external-cohort replication (AIBL [30], UK Biobank [31], OASIS-3 [32]) is explicitly named as the immediate next step (§4.2) rather than as a contribution. We do not run head-to-head comparisons against general-purpose splitting tools or biometric re-identification models; the contribution is the AD-specific framework plus the quantitative findings above.

2 Materials and Methods

2.1 A leakage taxonomy for slice-based AD MRI

We enumerate nine leakage and shortcut modes relevant to AD MRI deep learning. The list is intended as an operational checklist for AD MRI benchmarks rather than a closed taxonomy of every conceivable shortcut; the modes below are drawn from the prior leakage and shortcut literature [7, 8, 10, 13, 18, 20] and consolidated here into a structured form whose risk labels are calibrated to AD MRI deep-learning practice. Table 2 provides a structured overview. The new element relative to the existing leakage taxonomies is the explicit treatment of the near-duplicate (image-content) mode and the longitudinal-converter mode as first-class entries: both are characteristic of AD MRI workflows (augmented public releases, multi-visit research cohorts) and are not separately enumerated in general-purpose taxonomies such as Kapoor and Narayanan [20].

Subject, session, longitudinal, and slice leakage all violate the same clinical independence assumption: a test case should represent a new patient, not another view,

Table 2: Leakage and shortcut taxonomy for Alzheimer’s MRI deep learning. Risk levels are indicative for a redistributed public 2D benchmark and reflect the typical exposure of a CN-vs-AD slice-level classifier; absolute risk in any new study depends on cohort-specific structure.

Mode	Risk	Description
Subject leakage	Critical	Same patient in train and test.
Session leakage	High	Same visit/acquisition across splits.
Longitudinal leakage	High	Same patient at different time points.
Slice leakage	High	2D slices from one volume cross splits.
Near-duplicate	Moderate	Perceptually similar derived images.
Source leakage	Moderate	Dataset/collection provenance artifact.
Site/scanner	Moderate	Scanner, field strength, protocol artifact.
Demographic shortcut	Low–Mod	Age/sex imbalance correlated with label.
Clinical-label shortcut	Low	Advanced atrophy makes task trivially easy.

visit, or derivative of a patient already seen during training. This is especially important for 2D MRI pipelines because repeated slices from one 3D scan preserve head shape, intensity profile, slice position, and preprocessing traces that can function as biometric identifiers. Near-duplicate leakage is related but broader: visually similar derived images may cross splits even when filenames or metadata no longer expose their shared origin.

Source, site/scanner, and demographic shortcuts are not leakage in the narrow train-test contamination sense, but they can produce equally misleading clinical estimates. If label prevalence differs by dataset source, scanner protocol, age, sex, or disease severity, a model can learn cohort structure instead of disease-relevant neuroanatomy. Finally, clinical-label shortcut occurs when a benchmark mostly separates advanced atrophy from healthy controls; such performance may be real for that narrow severity contrast while still failing to support early diagnosis or deployment claims.

Figure 1 visualises the four leakage modes that share the train–test–contamination structure and that motivate the edge design of the SPLITGUARD-AD leakage graph in §2.2. The empirical decomposition in §3.4.1 shows that the dominant mode on the AD MRI cohorts studied here is longitudinal same-subject leakage (panel b), and that subject-only splitting alone (Protocol B) captures $\sim 93\%$ of the correction on the ADNI headline cohort. On Tier 1 that protocol is not available: grouping by the only identifier the release carries still leaves 70% of the test images with a participant seen in training (§2.3.1). The additional component-safe layer (Protocol C) contributes a smaller marginal above that baseline, because on cohorts with intact subject identifiers the two protocols produce the same partition. What separates them is what happens when those identifiers are not intact: §3.7 measures the divergence directly, and finds that Protocol C prevents 49% of the residual patient leakage Protocol B admits once a tenth of the identifiers are gone.

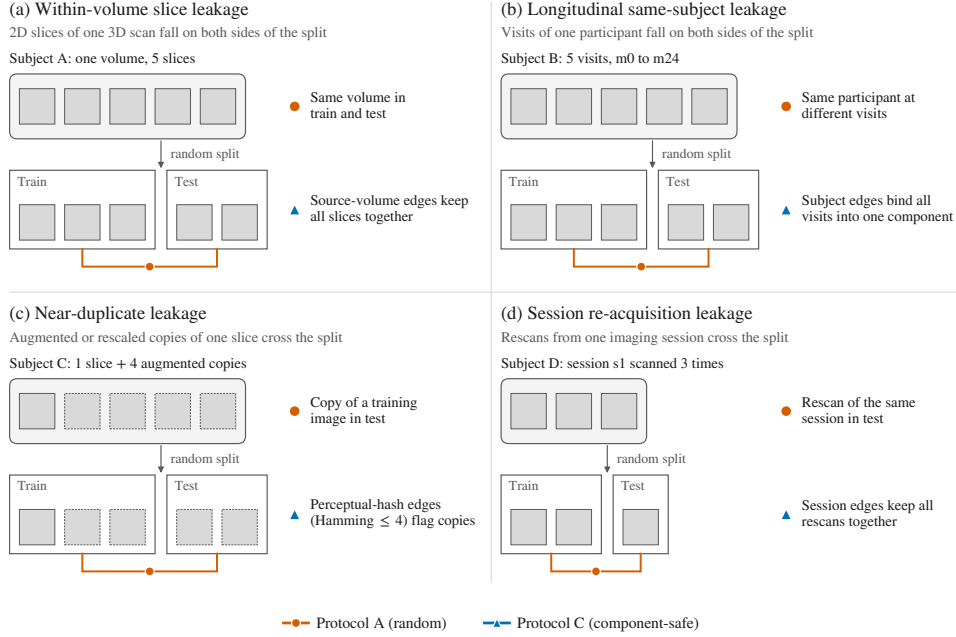


Fig. 1: Four leakage modes in slice-based AD MRI benchmarks, and the SPLITGUARD-AD edge type that addresses each: (a) source-volume (series-UID) edges, (b) subject-identifier edges, (c) near-duplicate dHash (Hamming ≤ 4 , reported not merged under the default policy), (d) session-identifier edges. In each panel the bracket and circle mark the leak a random split (Protocol A) produces, and the triangle the edge that prevents it under component-safe splitting (Protocol C). Connected components under the union of the blocking edges are the indivisible split units. **This figure describes the framework’s design space, not its empirical exercise in this paper.** On the two source cohorts every connected component contains exactly one participant; on Tier 1 the components built from the release’s own identifier contain up to three participants each, because that identifier is not a participant identifier (§2.3.1). Modes (a) and (d) do fire on ADNI (209 source-volume and 223 session pairs, contributing to 115 of the 382 components), but each joined a pair that mode (b) had already joined, so no component differs from a plain subject grouping; mode (c) is detected and reported but not merged under the default policy. All four are implemented and auditable; none is load-bearing on these cohorts, which is why the component layer’s measured marginal over subject-wise splitting is small (§3.4.1, Table 3). Demonstrating them requires a cohort with re-acquisition or augmentation structure that subject identity does not already capture; we name this as the primary validation gap in §4.2.

2.2 The audit instrument

SPLITGUARD-AD is an end-to-end evaluation framework comprising five sequential modules (Figure 2).

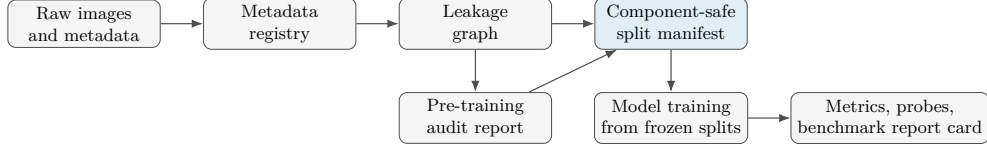


Fig. 2: The SPLITGUARD-AD pipeline: from raw images to audit-verified split manifests and benchmark report cards. Modules are executed sequentially; each produces a versioned artefact consumed by the next.

2.2.1 Metadata registry

Each image is assigned a row in a manifest with fields including `image_id`, `subject_id`, `subject_id_confidence`, `raw_class_label`, `binary_label`, `source_dataset`, and preprocessing lineage. For public JPEG benchmarks lacking original DICOM metadata, subject IDs are inferred from filename conventions with confidence flags for unparseable cases; Tier 1 shows that such inference can be wrong in ways no confidence flag detects (§2.3.1).

2.2.2 Leakage-graph construction

A leakage graph $G = (V, E)$ is constructed over the image set V . We distinguish the edge families that are *blocking* (those whose endpoints are merged into one component and therefore can never be split) from those that are *reported* for inspection, because the released implementation treats them differently and the distinction changes what the framework guarantees.

Blocking edges. An edge (u, v) is added, and the endpoints merged, if u and v share an inferred subject identifier, a session identifier, a source volume (identified by series UID), an acquisition date within a subject, or are byte-identical (equal SHA-256). Which of these are available is cohort-dependent, and the released audit reports the per-reason edge counts so a reader can see which rules actually fired: the ADNI tier supplies subject, session, series-UID, longitudinal and acquisition-date identifiers; the public-benchmark tier has no acquisition provenance at all, so only inferred subject and exact-SHA edges exist there. Table 3 gives the per-cohort breakdown.

The practical consequence is stated plainly: on the two source cohorts the connected components coincide with participant grouping, and on Tier 1 they coincide with an identifier that is not participant identity. The additional edge families are implemented, auditable and do fire, but none of them changes a single component here. Establishing their value requires a cohort whose leakage structure departs from subject identity (§4.2).

Reported edges. Near-duplicate pairs are detected with a difference hash (dHash) at 16×16 , i.e. a 256-bit code, and flagged when the Hamming distance satisfies $d \leq \delta$ with $\delta = 4$. Under the default policy (`--near-duplicate-policy review`) these pairs are written to a candidate file and surfaced in the audit report but are *not* merged; this is the policy under which every manifest and every number in this paper was produced. A **blocking** policy that additionally merges them is implemented and switchable. We keep review as the default because the identities recovered for Tier 1

Table 3: Which edge rules actually fire, per dataset, counted from the released graph-audit artefacts. The final column is the one that matters for interpreting the component layer: an edge is *load-bearing* only if it merges a pair that no other rule had already merged. On ADNI the session, series-UID and acquisition-date rules fire on 223, 209 and 222 pairs respectively, yet every resulting component still contains exactly one subject, so each of those edges joined a pair that `same_subject` had already joined. They are redundant on these cohorts rather than inert, and become load-bearing only where subject identity is absent, corrupted, or inferred wrongly. §3.7 tests that claim directly by deleting subject identifiers at controlled rates.

Dataset	Edge rule	Pairs joined	Load-bearing here?
ADNI1	<code>same_subject</code>	1,800	Yes (sole component-forming rule)
ADNI1	<code>same_session</code>	223	No (subsumed by subject)
ADNI1	<code>same_series_uid</code> (source volume)	209	No (subsumed by subject)
ADNI1	<code>same_acq_date</code>	222	No (subsumed by subject)
267/382 ADNI components are formed by <code>same_subject</code> alone; in the other 115 the redundant rules also fire.			
Kaggle	<code>same_subject</code> (filename-inferred)	6,076	Fires, but the key is not participant identity
Kaggle	<code>exact_sha256_duplicate</code>	0	No pairs found
Kaggle	<code>near-duplicate</code> (dHash ≤ 4)	1,582	Not merged (REVIEW policy)
OASIS-1	<code>same_subject</code>	all 416 components	Yes
OASIS-1	<code>same_session</code>	all 416 components	No (subsumed by subject)
OASIS-1	<code>same_source_volume</code>	all 416 components	No (subsumed by subject)

(§2.3.1) let this threshold be calibrated rather than assumed, and the calibration shows that no setting of it is safe to gate on. At $d \leq 4$ the detector is precise against recovered identity (precision 0.986) but sees only 1.6% of same-participant pairs, so merging its firings moves the Tier-1 test contamination from 70.0% to 61.2%: real, and nowhere near a fix. At $d \leq 8$, where it recovers 82% of adjacent-slice pairs, its 8,636 firings merge the 6,400 images into 27 components and join 1,343 pairs carrying opposite labels, so the split stops separating anything at all. A rule with that shape belongs in an audit report a human acts on, not in an automatic gate.

We report the hash family and width explicitly because they are not interchangeable: dHash at 256 bits with $\delta = 4$ is a far stricter criterion than the 64-bit perceptual hashes for which the $\delta = 4$ convention was originally proposed. Connected components $C_1, \dots, C_k$ of G define the atomic units for splitting: no component may be split across train, validation, and test partitions.

2.2.3 Component-safe split generation

Given target split ratios $r_{\text{train}} : r_{\text{val}} : r_{\text{test}}$ and a class stratification target, SPLITGUARD-AD assigns entire components to splits using a greedy bin-packing procedure seeded for reproducibility. The resulting split manifest is frozen as a CSV file and versioned; no training script may discover or modify splits at runtime. Algorithm 1 formalises the full procedure: leakage graph construction (lines 1–8) followed

by component-safe assignment (lines 9–18). All randomness is funnelled through the single SEED argument, so the manifest is byte-reproducible.

Algorithm 1 SPLITGUARD-AD leakage-graph construction and component-safe split generation.

Require: image set V ; subject-ID map σ ; session-ID map ς ; source-volume (series-UID) map ν , each defined only where the cohort supplies it; dHash distance $d(\cdot, \cdot)$ (256-bit code) with threshold $\delta = 4$; near-duplicate policy $\rho \in \{\text{REVIEW}, \text{BLOCKING}\}$ (default REVIEW; see §2.2.2); target ratios $(r_{\text{tr}}, r_{\text{val}}, r_{\text{te}})$; class label $y : V \rightarrow \mathcal{Y}$; RNG seed s

Ensure: split assignment $\pi : V \rightarrow \{\text{train}, \text{val}, \text{test}\}$ such that no leakage component is split across partitions

```

1:  $E \leftarrow \emptyset$ ;  $R \leftarrow \emptyset$  ▷  $E$ : blocking edges;  $R$ : reported-only pairs
2: for each pair  $(u, v) \in V \times V$  with  $u \neq v$  do
3:   if  $\sigma(u) = \sigma(v)$  or  $\varsigma(u) = \varsigma(v)$  or  $\nu(u) = \nu(v)$  then
4:      $E \leftarrow E \cup \{(u, v)\}$  ▷ subject, session, or source-volume edge, always blocking
5:   else if  $d(u, v) \leq \delta$  then
6:     if  $\rho = \text{BLOCKING}$  then
7:        $E \leftarrow E \cup \{(u, v)\}$ 
8:     else
9:        $R \leftarrow R \cup \{(u, v)\}$  ▷ surfaced in the audit report, not merged
10:    end if
11:  end if
12: end for
13:  $\{C_1, \dots, C_k\} \leftarrow$  connected components of  $G = (V, E)$ 
14: // Bin-pack components into partitions under class stratification
15: Initialise empty bins  $B_{\text{tr}}, B_{\text{val}}, B_{\text{te}}$ 
16:  $\text{rng} \leftarrow \text{RANDOMGENERATOR}(s)$ 
17: Sort components by  $|C_i|$  descending; tie-break by class-balance deficit
18: for  $C_i$  in sorted order do
19:   // Deficit for bin  $B$ , component  $C_i$ , target ratios  $r$ :
20:   //  $\text{deficit}(B, C_i, r) = \sum_{\text{class } y} \left| \frac{|B \cup C_i|_y}{|V| \cdot r_B} - \frac{|V|_y}{|V|} \right|$ ,
21:   // i.e. sum over classes of the absolute deviation from the cohort-wide class
   share, normalised to bin  $B$ 's target ratio  $r_B$ .
22:    $B^* \leftarrow \arg \min_{B \in \{B_{\text{tr}}, B_{\text{val}}, B_{\text{te}}\}} \text{deficit}(B, C_i, r)$ 
23:    $B^* \leftarrow B^* \cup C_i$ 
24: end for
25:  $\pi \leftarrow$  assignment induced by  $(B_{\text{tr}}, B_{\text{val}}, B_{\text{te}})$ 
26: return  $\pi$ , audit report {component sizes, class balance, cross-partition contamination, source/site/demographic associations}

```

Implementation note

The double loop on lines 2–6 of Algorithm 1 is shown in $\mathcal{O}(N^2)$ form for clarity; in implementation, exact identifier edges (subject, session, series UID, acquisition date,

SHA-256) use hash-grouping in $\mathcal{O}(N)$. The near-duplicate comparison is the exception: it is currently evaluated over all $\binom{N}{2}$ pairs, which is tractable at the scale studied here ($N \leq 6,400$) but is the component that would need locality-sensitive indexing before the framework is applied to repositories of 10^5 images or more. Identifier-edge construction completes in < 1 ms per cohort on a consumer CPU, so framework runtime is dominated by image ingestion and by the pairwise hash comparison.

Protocol nomenclature

Throughout the experiments we evaluate three split protocols on the same image universe:

Protocol A (Random, leaky). Random image-level shuffle into train/val/test; the conventional benchmark setup.

Protocol B (Subject-only). Split by inferred subject identity: all images from one subject land in a single partition, but connected-component and near-duplicate safety are *not* enforced. When subject identity is unavailable, or when leakage components coincide with subject boundaries, Protocol B is not separately informative from Protocol C; we therefore report it only where it isolates a distinct safety layer (the ADNI experiments in §3.3).

Protocol C (Component-safe, SPLITGUARD-AD). Full leakage-graph component-safe split: subject identity, scanner-session re-acquisitions, and near-duplicate components are all respected.

The three protocols form a strict ordering ($A \supseteq B \supseteq C$) in terms of leakage permissiveness; the three-point degradation analysis in §3.4.1 traces the contribution of each layer.

2.2.4 Pre-training audit

Prior to model training, SPLITGUARD-AD produces a structured audit report verifying: (i) zero component overlap across splits; (ii) label distribution per split; (iii) source/site/demographic association with label, reported as a per-axis Cramér’s V between the audited covariate (site code extracted from ADNI PTID, scanner field strength, sex, age band) and the binary CN/AD label, computed per partition (a complementary axis to scanner-effect work such as Glocker *et al.* [33] and to harmonisation methods in the ComBat lineage [34–36]; SPLITGUARD-AD surfaces the confound but does not attempt to remove it, because the leakage gap this paper measures is dominated by the subject-identity channel that harmonisation does not fix post-hoc); and (iv) a clinical claim boundary statement. The audit report issues a binary GO / NO-GO decision for training according to the formal criteria in Table 4.

2.2.5 Model audit layer

After training, SPLITGUARD-AD evaluates trained model representations with a linear probe: probe accuracy above chance indicates residual shortcut encoding. In this release only the *subject-identity* probe is implemented and reported (§3.4, `scripts/run_biometric_probe_adni.py`), because subject identity is the channel

Table 4: GO/NO-GO decision criteria of the released SPLITGUARD-AD audit (`scripts/audit_adni_splits.py`). We distinguish what the released gate *enforces* from what it *reports* and from what the framework specifies but does not yet enforce, because a leakage audit whose advertised criteria exceed its implemented criteria is itself an evaluation-integrity failure. Only the three blocking checks determine the binary decision; every other quantity is written to the audit report for author and reviewer judgement without halting the pipeline. The gate certifies partition disjointness of the split manifest, *not* site-generalisability, scanner-shift robustness, calibration, fairness, or clinical deployment readiness.

Check	Status in v1	Behaviour
Subject overlap across partitions	Blocking	Any subject ID in > 1 partition sets passed=False.
Session overlap across partitions	Blocking	Any session ID in > 1 partition sets passed=False.
Component overlap across partitions	Blocking	Any component ID in > 1 partition sets passed=False.
Per-partition label balance	Reported	Class counts and ratios written to the report.
Scanner field-strength balance	Reported	Marginal distribution per partition.
Age and sex summaries	Reported	Per-partition demographic summary.
Site-label association (Cramér’s V)	Reported	Computed by a separate script (§3.3); not consulted by the gate.
Component-size balance across partitions	Reported	Per-partition component-size distribution, with a warning when partitions are not drawn from comparable participants. Added after auditing this project’s own frozen splits, on which it fires (§4.2); not consulted by the gate.
Unresolved subject or diagnosis identifiers	Not enforced in v1	No check exists. Two ADNI scans with unresolved visit-level diagnosis passed the gate for this reason (§2.3.3).
Frozen-manifest hash verification	Not enforced in v1	Hashed manifests are produced for release but the gate does not re-verify them.
Residual near-duplicate cross-split	Not enforced in v1	Near-duplicate edges are applied when <i>building</i> components, not re-checked after partitioning.
Within-component label agreement	Not enforced in v1	Mixed-label converter components are handled by an explicit pre-specified exclusion rule (§2.3.3) rather than by the gate.

the component-safe split is designed to close. Probes for dataset source, site/scanner, and demographic attributes follow the same construction and are the natural extension of this layer, but are not implemented in v1 and no such probe result is reported here; the site-label association we do report (§3.3) is a metadata-level contingency statistic, not a representation-level probe.

2.3 Cohorts

Cohort roles

Two source cohorts carry the measurements. The **ADNI1: Complete 3Yr 1.5T** collection (§2.3.3) is the paper’s *headline* cohort: all headline numerical claims in the abstract, the cost-of-leakage analysis, the subgroup analysis, and the dose-response stress test derive from it. **OASIS-1** (§2.3.2) is the metadata-rich replication cohort, where true participant identifiers are authoritative; it is operationally external to

ADNI (different recruiting site at Washington University, different scanner and acquisition protocols, different recruitment era), so the direction-preservation we report between ADNI and OASIS-1 functions as a partial cross-site generalisation check even though we do not describe it as “external validation” in the strict regulatory sense. The third tier is not a third cohort. The **Kaggle public JPEG benchmark** (§2.3.1) is, as we show by content matching, a redistribution of OASIS-1 itself whose participant identifiers have been replaced by a filename index that both merges and splits participants. We therefore use it for what it documents rather than for what it would replicate: an observed and measurable instance of the provenance damage that §3.7 otherwise has to simulate. Throughout the experiments we use the shorthand *Tier 1* (redistributed public benchmark), *Tier 2* (OASIS-1), *Tier 3* (ADNI1 headline) for brevity.

2.3.1 Tier 1: a redistributed public benchmark

The widely distributed public Kaggle Alzheimer’s MRI benchmark dataset [37] comprises 6,400 JPEG images of uniform 176×208 pixels in four severity classes: NonDemented ($n = 3,200$), VeryMildDemented ($n = 2,240$), MildDemented ($n = 896$), and ModerateDemented ($n = 64$). The release states no origin, carries no acquisition metadata, and exposes subject identity only through a parenthesised filename index (*e.g.*, `NonDemented_patient100_1 (100).jpg`).

What the images are

We recovered the origin by content. Every image was matched against axial slices of all 436 OASIS-1 cross-sectional volumes in their processed, atlas-registered, masked form (`T88_111/*_masked_gfc; scripts/audit_tier1_ground_truth.py`). All 6,400 images match unambiguously: the lowest correlation between an image and its matched slice is 0.9996, while the highest correlation any image reaches with a slice of a *different* participant is 0.9623. The benchmark is 200 OASIS-1 participants at 32 consecutive axial slices each (atlas slices 71–102). The folder labels reproduce those participants’ Clinical Dementia Rating exactly, for every one of the 6,400 images ($\text{CDR} = 0 \rightarrow \text{NonDemented}$, $0.5 \rightarrow \text{VeryMild}$, $1 \rightarrow \text{Mild}$, $2 \rightarrow \text{Moderate}$), and no participant’s slices are split across two folders: they are CDR, not the weak folder-derived labels they are usually assumed to be. Every one of the 200 participants is also in our Tier-2 OASIS-1 set, so Tier 1 and Tier 2 are two views of one cohort, and the cross-tier replication in this paper rests on two source cohorts rather than three.

What redistribution destroyed

The filename index is not a participant identifier. Its 324 distinct keys divide into 149 that correspond to exactly one participant and 175 that merge two or three participants under a single key, while 179 of the 200 participants are spread across two or more keys and one is spread across 32. A further 128 images (2.0%) carry no parsable index at all. Grouping by this key is therefore not subject-wise splitting: it is grouping on a key that is wrong in both directions at once, and nothing inside the release reveals that.

Against the recovered identities the damage is exact. In the frozen seed-42 split used throughout this paper, which groups by the filename key, 85 participants have images in more than one partition, 46 of the 58 test participants also appear in training, and 647 of the 960 test images (67.4%) belong to a participant the model was trained on. Table 5 repeats that measurement for every grouping rule the release supports.

Table 5: Every grouping rule the Tier-1 release supports, measured against the recovered participant identities over five seeds. Contamination is the share of test images whose participant also appears in training. The filename key, the only identifier the release provides, leaves seven test images in ten contaminated. Near-duplicate edges at $d \leq 4$ remove an eighth of that; at $d \leq 8$ the graph collapses (8,636 pairs merge the corpus into 27 components, the largest holding 3,117 images, and 1,343 of the merged pairs join images with opposite labels), which is a different failure rather than a fix. Only the recovered participant identity yields a clean split.

Grouping rule	Components	Largest (images)	Test contamination mean [min, max]	Test participants in train / total
Filename key (release identifier)	324	32	70.0% [62.0, 78.6]	46.4/59.4
+ near-duplicate $d \leq 4$	219	192	61.2% [53.4, 67.5]	42.0/56.2
+ near-duplicate $d \leq 8$	27	3,117	100%	1.0/1.0
Near-duplicate $d \leq 8$ alone	999	1,770	99.5% [99.0, 100]	104.4/105.0
Recovered participant	200	32	0%	0/29.0

Near-duplicate structure, read against ground truth

No exact SHA-256 duplicates exist. The near-duplicate detector (dHash at 16×16 , a 256-bit code, Hamming $d \leq 4$) flags 1,582 candidate pairs, of which 1,559 (98.5%) join two slices of the same participant and 1,526 join slices one position apart in the same volume. Of the 57 pairs that straddle the frozen partitions, the recovered identities make 52 the same participant and 5 different participants. The filename key had called 36 of those 57 pairs “different subjects”; in 31 of them the key was wrong and the detector was right. Read against the release’s own identifier the detector looks like a 63% false-positive rate; read against ground truth its cross-partition output is 91% true leakage. Only the recovered provenance distinguishes the two readings, which is the point: an audit that can only compare a detector against the identifiers it already has will systematically mistake a broken identifier for a false alarm.

Calibrating the detector

With every pair’s identity known, the threshold can be calibrated over all 20,476,800 pairs instead of argued for. At $d \leq 4$ the detector is precise and sparse: precision 0.986 and recall 0.246 of adjacent-slice pairs, which is 0.016 of all pairs sharing a participant, at a false-positive rate of 1.1×10^{-6} . At $d \leq 8$ it is sensitive and unusable as a blocking rule: precision 0.699, recall 0.816 of adjacent slices, false-positive rate 1.3×10^{-4} , which at this corpus size is the 8,636 pairs that collapse the graph in Table 5. The median Hamming distance between two slices of the same participant is

33 bits against 35 between participants, so what the detector separates is *neighbouring slices*, not participants. This is the precise sense in which perceptual hashing cannot stand in for a participant identifier.

2.3.2 Tier 2: OASIS-1

We use OASIS-1 as the metadata-rich replication cohort [38]; LaMontagne *et al.* [32] describe the successor OASIS-3 longitudinal release for context. All 12 official raw cross-sectional archives were available locally, together with the demographic/clinical and reliability spreadsheets. The clinical spreadsheet contains 436 session rows from 416 unique subjects, including 20 MR2 reliability sessions. For each session, SPLITGUARD-AD selectively extracts one processed masked T88 MPRAGE Analyze pair (PROCESSED/MPRAGE/T88_111/*_masked_gfc.img/.hdr) without fully expanding the raw archives.

CDR is used as the primary clinical label: $CDR = 0$ is mapped to NonDemented, and $CDR > 0$ to Demented. Sessions with missing CDR are retained in the manifest for auditability but excluded from the primary classification experiment. The resulting OASIS-1 manifest contains 135 NonDemented sessions, 100 Demented sessions, and 201 sessions with missing CDR. Five central coronal slices are derived from each processed volume, yielding 2,180 total slices and 1,175 labelled slices (675 NonDemented, 500 Demented). The frozen OASIS-1 SPLITGUARD-AD split contains 825/175/175 labelled slices in train/validation/test, with 165/35/35 subjects respectively and zero subject or component overlap across splits.

2.3.3 Tier 3: ADNI1 Complete 3Yr 1.5T (headline cohort)

We use the ADNI1: Complete 3Yr 1.5T image collection as the paper’s headline cohort, accessed under an active LONI IDA Data Use Agreement; the consortium overview [39], the MRI-acquisition protocol [40], and the clinical-characterisation core [41] provide the foundational methodology references for the ADNI release used here. All headline numerical claims in the abstract, the intro, and the cost-of-leakage and dose-response analyses derive from this cohort. The collection contains 2,182 pre-processed 1.5T T1-weighted volumes (MPR or MPR-R variants with GradWarp, B1 correction, N3, and intensity scaling already applied) from 382 unique subjects spanning the ADNI1/GO/2/3/4 phases (Table 6). A small subset (28 scans, 17 of which fall in the CN/AD universe) are magnetization-transfer T1 (MT1) acquisitions rather than MPRAGE; these are retained in the primary analysis and excluded in a pre-specified sensitivity arm (§3.3).

Diagnosis labels are taken from the ADNI diagnostic summary table (DXSUM_PDXCONV_ADNIALL) under a phase-aware visit-level mapping. ADNI2–ADNI4 visits use the canonical DIAGNOSIS code (1=CN, 2=MCI, 3=AD) with longitudinal DXCHANGE transitions collapsed to the destination state. ADNI1 visits, which lack a single DIAGNOSIS code in this export, are resolved through the mutually exclusive legacy flags DXAD, DXMCI, DXNORM (the ADNI1 case report form encodes diagnosis across three Boolean indicators rather than one categorical field). Scan-to-visit alignment uses a 180-day date-proximity join through the MRI metadata table, with a documented date-only fallback when the viscode-keyed lookup fails. The full mapping

Table 6: ADNI1: Complete 3Yr 1.5T cohort demographics. Subject-level summary; subjects are assigned to a diagnosis group by highest-priority label observed across all visits (AD > CN > MCI). Age computed at each subject’s earliest scan from PTDOBY; baseline MMSE is each subject’s earliest non-missing MMSCORE.

	CN	MCI	AD	Total
Subjects (<i>n</i>)	142	68	172	382
Scans (<i>n</i>)	794	442	946	2182
Age, years (mean $\pm$ SD)	75.7 $\pm$ 5.4	75.1 $\pm$ 7.1	75.0 $\pm$ 7.3	n/a
Female (<i>n</i> , %)	68 (48%)	15 (22%)	76 (44%)	n/a
Baseline MMSE (mean $\pm$ SD)	29.2 $\pm$ 1.0	27.5 $\pm$ 1.7	24.7 $\pm$ 2.5	n/a

rule and edge-case handling are pre-specified before any model training; the resulting manifest yields 756 CN, 768 MCI, and 656 AD scans, plus two scans whose diagnosis cannot be resolved (kept in the manifest with `diagnosis_group=unknown` for auditability and excluded from the primary CN-vs-AD task).

Linkage-integrity audit

Because every ADNI label in this study rests on this join, we account for it per scan in `adni_linkage_audit.csv` (`scripts/build_linkage_audit.py`; schema in Supplementary Material S5). The table is an ADNI-derived Tier-3 artefact and is therefore not redistributed with the release; the LONI data-use agreement permits only hashed derivatives, which a reader with ADNI access regenerates from their own download by running that script. Each row records the scan identifier, a salted PBKDF2 derivation of the participant identifier, the acquisition date, the join tier, the source column, and whether the scan enters the CN-vs-AD universe. Of the 2,182 scans, 1,695 (77.7%) were labelled through a direct (RID, VISCODE) match, 485 (22.2%) required the 180-day date-proximity fallback, and 2 (0.1%) remained unresolved; these last two are the `diagnosis_group=unknown` scans discussed above. That roughly one label in five depends on a date match rather than an exact visit key is a real limitation of the label chain and we prefer to state it than to leave it implicit. Participant identifiers are hashed under a secret per-release salt for the same reason as the Tier-3 manifest: ADNI PTIDs are low-entropy, so a published salt would make the file re-identifiable.

For apples-to-apples comparison with Tier 2, each volume is reduced to a single 2D coronal-centre slice. The slice is computed once at ingest with 1st–99th percentile intensity normalisation and saved as a single-channel PNG; this caching step is byte-identical to the volume-time computation in §2.3.2 but fits the full collection in 54 MB rather than 77 GB of NIFTI volumes. The leakage graph links scans sharing a subject identifier, a session identifier, a series UID, an MRI acquisition pipeline, or appearing across longitudinal follow-up visits; the 382-subject collection collapses into 382 single-subject components (mean 5.7 scans per component, range 4–11), so the frozen SPLITGUARD-AD split is effectively a subject-stratified partition with additional safety against near-duplicate and series-level edges.

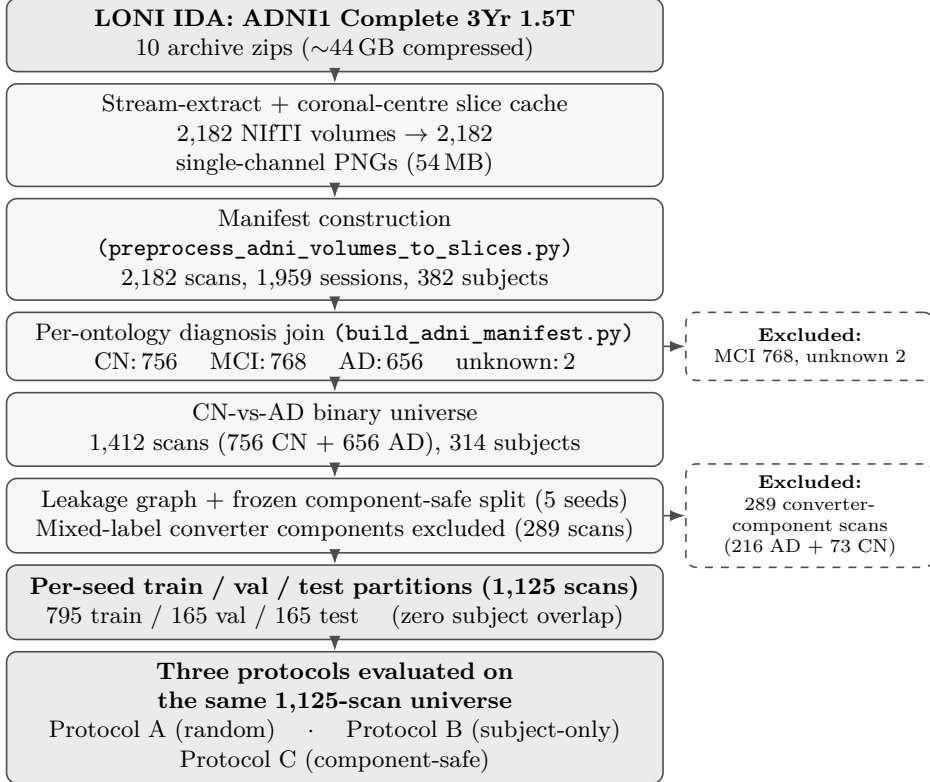


Fig. 3: ADNI1: Complete 3Yr 1.5T data flow from LONI IDA archive download to per-seed train/validation/test partitions. Each transition annotates the row count and the script responsible (`preprocess_adni_volumes_to_slices.py`, `build_adni_manifest.py`, `make_adni_splitguard_split.py`). Dashed boxes on the right record exclusions: MCI / unknown diagnoses excluded from the CN-vs-AD task by construction, and 289 CN/AD scans (216 AD + 73 CN) from mixed-label longitudinal converter components excluded from the primary arm and re-included in the sensitivity arm (§3.3).

The CN-vs-AD binary task universe excludes 768 MCI scans by construction (the binary task is cross-sectional, not longitudinal progression-prediction) and a further 289 scans (216 AD + 73 CN) drawn from components tagged `mixed_label`: subjects whose visit sequence crosses CN, MCI, and AD diagnoses (longitudinal converters). Including those subjects in the primary arm would require either (i) breaking the subject-level partition (the same subject contributes CN visits to one split and AD visits to another), or (ii) collapsing their longitudinal trajectory to a single subject-level label, which conflates cross-sectional CN-vs-AD classification with progression-prediction. We adopt the cleaner option: exclude these 289 scans from the primary arm and re-include them in a pre-specified sensitivity arm (§3.3, “Sensitivity to converter exclusion”), reclassifying each mixed-label component to its

CN-or-AD majority. The total inflation gap in the sensitivity arm (+0.142) is *larger* than in the primary arm (+0.118), so the exclusion does not drive the headline result (Table 11). Removing these 289 scans leaves 1,123 CN/AD scans. The frozen partitions contain 1,125 scans, because two scans whose visit-level diagnosis did not resolve (`diagnosis_group=unknown`, §2.3.3) were retained under the CN/AD label inherited from their single-subject component rather than dropped. Both are assigned to the training partition in all five seeds and therefore enter no reported test metric; we record them here rather than silently reconciling the count, and the released manifests flag them explicitly. The primary CN-vs-AD universe is thus 1,123 diagnosis-resolved scans plus these two, across 220 single-subject components, equivalently 220 participants, since every component here contains exactly one subject; each frozen split produces train/validation/test partitions of 795/165/165 with zero subject or component overlap across partitions. Figure 3 summarises the end-to-end pipeline from the 10 LONI IDA archive zips through the slice-cached manifest to the per-seed splits.

2.4 Use of large-language-model tools

The journal asks that any use of a large language model (LLM) be documented in the Methods, so we state it here rather than only in the Declarations. LLM-based coding assistants were used throughout the implementation of the released software: the training, audit, split-generation and analysis scripts, the orchestration and verification tooling, and the figure and table generators were written and revised with such assistance, as were drafts of this manuscript’s prose. They were also used to find defects, one of which is worth naming because it is the kind of error this paper is about: the volumetric trainer computed $1 - \text{AUROC}$ from a descending rank sum, which would have selected the least-trained epoch of every volumetric run. That arm is not reported here (§4.2), and the released code carries a regression test pinning the trainer’s metric to the analysis pipeline’s.

What the tools did not do is decide anything. The study design, the protocol definitions, the choice of cohorts and arms, the pre-specification of the robustness arms, and the interpretation of every result are the authors’. No number in this manuscript was produced or altered by an LLM: every reported quantity is computed by the released scripts from the released manifests, and `scripts/verify_paper_numbers.py` re-checks each of them against the artefact it comes from, so a value that an LLM had introduced into the text would fail that check rather than survive it. No text, data or citation was accepted without author verification, and the authors take full responsibility for the content of this publication.

3 Results

Statistical conventions

Unless stated otherwise, intervals are $B = 10,000$ paired-seed bootstrap intervals [42, 43] over a five-seed array. At $n = 5$ they describe seed-level variability rather than population-level uncertainty, and the “95%” label is kept only for tabulation convention. Where we say a direction holds, we mean the sign of the inter-protocol delta is the same in every one of the five seeds; we do not quote the resample count, because

resampling five same-signed numbers reproduces that sign with probability one and so adds nothing to the seed count itself. A two-level hierarchical (subject $\times$ seed) bootstrap is run on every arm from its per-image predictions and reported alongside; it is the more defensible inferential quantity and is roughly $1.9\text{--}4.7\times$ wider across the arms. We weight, in decreasing order, effect size, sign consistency, the hierarchical interval, and last the paired-seed width.

Training recipe and compute

All cohorts and both backbones share one recipe: ImageNet-pretrained ResNet-18 (or DenseNet-121 in the architecture arms), $3\times 224\times 224$ input with ImageNet mean/std normalisation, horizontal flip and $\pm 7^\circ$ rotation augmentation, AdamW with cosine annealing, weight decay 10^{-2} , class-weighted BCE loss and one-shot test-set evaluation. Cohort-specific settings are learning rate 3×10^{-4} (Tier 1, OASIS-1) or 10^{-4} (ADNI); 30, 20 and 15 epochs; batch size 32 on Tier 1 and OASIS-1 and 16 on ADNI; seeds $\{42, 0, 1, 2, 3\}$ on Tier 1 and OASIS-1 and $\{0, 1, 2, 3, 4\}$ on ADNI. The volumetric arm uses a 3D ResNet-18 on 128^3 volumes, batch size 4, 15 epochs. Every run was executed in one code state with deterministic cuDNN kernels and seeded data loaders, on one NVIDIA H100 except the dose-response matrix, rerun on an NVIDIA RTX PRO 4500, and the volumetric arm, run on an NVIDIA RTX 2000 Ada (§Declarations); the full configuration is in Supplementary Table S1.

3.1 Observed provenance damage on a redistributed benchmark

Setup

Tier 1 is trained under three protocols that differ only in what they group by. Protocol A shuffles images at random. Protocol B' groups by the filename key, the only identifier the release carries and therefore the strongest protocol an audit run from inside the release can implement. Protocol C' groups by the participant identity recovered in §2.3.1. Everything else is held fixed: ResNet-18 [44] with ImageNet pre-trained weights, learning rate 3×10^{-4} , weight decay 10^{-2} , batch size 32, 30 epochs, random horizontal flip and $\pm 7^\circ$ rotation, AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, AMS-Grad disabled) with cosine annealing, class-frequency-weighted BCE loss, and the seed array $\{42, 0, 1, 2, 3\}$.

Leakage verification

Under Protocol A, 195 of 219 test-set filename keys (89.0%) also appear in training. Under Protocol B' no filename key crosses a partition, so an audit restricted to the release's own metadata reports zero leakage, while 70.0% of the test images in fact belong to a participant the model trained on. Under Protocol C' the contamination is zero (Table 5).

Results

Random splitting inflates test AUROC from 0.766 to 0.999, a gap of +0.233. That part is the familiar result. The part this tier exists for is the middle column: grouping

Table 7: Tier-1 test AUROC under the three grouping rules, five seeds. Protocol B' groups by the filename key the release ships, which is the best an audit inside the release can do; Protocol C' groups by the participant identity recovered from OASIS-1. The gap column is A-C'.

Seed	A: random	B': filename key	C': participant	Gap (A-C')
42	0.9994	0.8581	0.8387	+0.161
0	0.9999	0.7508	0.7655	+0.234
1	0.9996	0.8142	0.7496	+0.250
2	0.9976	0.7834	0.7743	+0.223
3	0.9999	0.7574	0.7030	+0.297
Mean $\pm$ SD	0.999 $\pm$ 0.001	0.793 $\pm$ 0.044	0.766 $\pm$ 0.049	+0.233

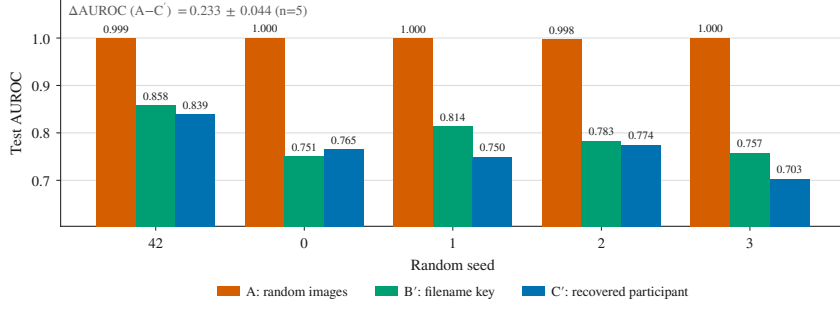
by the release’s own identifier recovers 88.6% of that gap and leaves +0.027 AUROC of inflation standing, because the key it groups on merges and splits participants. A practitioner who does everything correctly with the information the benchmark provides still reports an inflated number, and nothing inside the benchmark reveals it. This is the difference between leakage as a protocol error, which the literature already documents, and leakage as a provenance failure, which survives a correctly applied protocol. Figure 4 separates what is stable across seeds from what is not. Protocol A sits above both grouped protocols in all five seeds (Figure 4a) and on every metric at seed = 42 (Figure 4b), so the headline gap is not a seed artefact. The residual between B' and C' is the smaller quantity and it is not uniform: their AUROC ordering reverses in one seed of five (Table 7), and at seed = 42 C' is the more specific of the two while B' is the more sensitive, which is what a threshold applied to a less separable score produces. The +0.027 residual is therefore a five-seed mean and not a per-run guarantee; what the mechanism predicts, and what the data supports, is that grouping on a broken key leaves inflation standing on average, not that it leaves more in every run.

What the three test sets are not

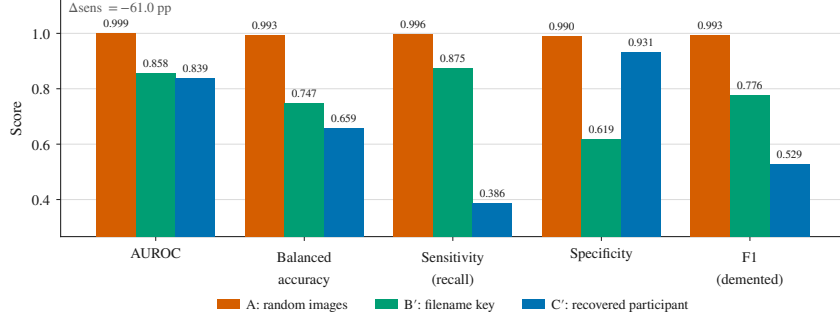
A random split spreads a participant’s 32 slices over all three partitions, so its test partition holds many more participants, each contributing far fewer images, than the participant-grouped one. That asymmetry is a property of the protocols and cannot be removed by resampling: subsampling participants leaves AUROC essentially unchanged, because it is a rank statistic over pairs. What it does mean is that the three AUROCs are estimated over differently composed test sets, and we report the composition rather than adjusting for it (§3.3, Table 12).

Learning curves

Figure 5 shows validation AUROC per epoch. Protocol A saturates within the first third of training, which is the signature of memorising images it will be tested on rather than learning a disease feature; Protocol B' tracks it at a lower level, still lifted by the participants its key failed to group; Protocol C' converges lowest and slowest.



(a) Per-seed test AUROC under the three grouping rules.



(b) All test-set metrics at seed = 42.

Fig. 4: Tier-1 under random splitting (A, vermilion), the release’s filename key (B’, green) and recovered participant identity (C’, blue): (a) seed stability, and (b) the per-metric breakdown. Panels are stacked vertically rather than side-by-side so that the in-figure tick labels and bar annotations remain legible at print size.

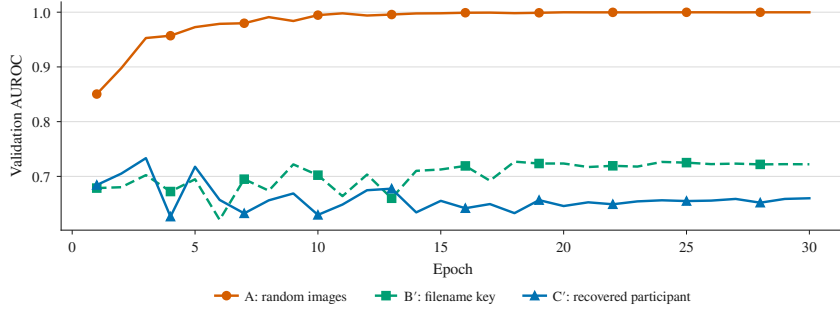


Fig. 5: Validation AUROC per epoch at seed = 42 for Protocol A (random, vermilion), Protocol B’ (filename key, green) and Protocol C’ (recovered participant, blue). The vertical distance between the first and last curves is the inflation the release’s own identifier cannot remove.

Tier-1 takeaway

Tier 1 is not an independent cohort and we no longer treat it as one: it is 200 OASIS-1 participants redistributed without their identifiers. That makes it the one place in this paper where provenance damage is observed rather than simulated, and it supplies the quantity the simulation in §3.7 is calibrated against: an identifier that is wrong in both directions leaves +0.027 AUROC of inflation in place after a correctly applied subject-wise split. The mechanism behind that inflation is examined in §3.4, the layer-by-layer decomposition in §3.4.1, and the cohorts with intact identifiers in §3.2 (OASIS-1) and §3.3 (ADNI).

3.2 Replication on OASIS-1

We repeat the leaky-versus-SPLITGUARD-AD comparison on OASIS-1 using the same ResNet-18 family and the derived 2D slice representation described in §2.3.2. Protocol A randomly splits the 1,175 labelled slices by image; Protocol C uses the frozen subject-component-safe SPLITGUARD-AD split. Across five random seeds, Protocol A places nearly every test subject in training (mean test-subject contamination $99.54\% \pm 0.68\%$, range 98.5–100%).

Why the OASIS-1 gap is mechanically smaller than ADNI’s

Before we report the AUROC numbers, we surface the structural properties of the OASIS-1 slice manifest that bound the leakage surface *a priori* (script: `scripts/oasis_subject_level_summary.py`; JSON: `reports/tables/oasis1_subject_level_summary.json`). Across the labelled subset ($n = 235$ subjects with binary NonDemented/Demented labels; a further 181 subjects are dropped for missing CDR at the labelling stage) OASIS-1 has 5 coronal slices per session (median = mean = 5; range 5–5), ~ 1.05 sessions per subject on average (only 20 subjects, 4.8% of the 416 imaged subjects, carry a repeat session), and a total of 2,180 slices spread across 436 sessions. Per-subject image counts are in fact comparable between the two cohorts (OASIS-1 ≈ 5.2 slices per subject; ADNI1 primary arm ≈ 5.1 scans per subject at one coronal-centre slice per scan, §2.3.3), so the smaller OASIS-1 gap is not attributable to a thinner per-subject image stack. The more plausible mechanism is structural: only 4.8% of OASIS-1 subjects carry a repeat session, so the cohort contains almost no longitudinal within-subject variation for a model to exploit, whereas the ADNI1 cohort is longitudinal by construction. The OASIS-1 replication is therefore best read as a directional replication of the mechanism, not as a high-powered replication of the inflation magnitude reported on ADNI (§3.3). The OASIS-1 gap is reported at the image level, and should be read as the weaker of the two cross-cohort quantities when quoted alongside the ADNI subject-level result.

The OASIS-1 result (Table 8) validates the core leakage mechanism on the cohort Tier 1 was drawn from: the leaky protocol still places nearly every test subject in the training set (99.54%). While the operating-point metrics (F1, sensitivity) are statistically noisy across five seeds due to the smaller dataset size, and one seed produces a negative AUROC gap, the contamination result confirms that random slice splitting creates severe subject leakage even when true metadata are available.

Table 8: OASIS-1 replication: random slice split against the participant-safe SPLITGUARD-AD split, mean $\pm$ SD over five ResNet-18 seeds at 20 epochs.

Metric	Protocol A	Protocol C	Gap (A–C)
AUROC	0.828 ± 0.029	0.793 ± 0.020	0.035 ± 0.037
Balanced accuracy	0.712 ± 0.027	0.704 ± 0.024	0.008 ± 0.043
F1 (demented)	0.650 ± 0.069	0.683 ± 0.015	-0.033 ± 0.076
Sensitivity	0.687 ± 0.225	0.771 ± 0.137	-0.084 ± 0.220
Specificity	0.738 ± 0.174	0.638 ± 0.179	0.100 ± 0.185
Test subjects leaking	99.54%	0%	n/a

DenseNet-121 architecture replication

Under a DenseNet-121 backbone and the same five seeds the OASIS-1 gap is +0.048, mirroring the ResNet-18 result above in point estimate and in seed-to-seed noise on this small cross-sectional cohort.

3.3 Replication on ADNI1, and its robustness arms

We repeat the leaky-versus-SPLITGUARD-AD comparison on the ADNI cohort introduced in §2.3.3, using the same ResNet-18 family and coronal-centre-slice representation as §3.2. Three split protocols are compared on the same 220-participant CN-vs-AD manifest (1,125 rows, of which 1,123 are diagnosis-resolved; partition sizes differ slightly by protocol: 169/169/165 test scans for Protocols A, B and C respectively, because the component-safe bin-packing cannot hit the target size exactly): Protocol A randomly splits images 70/15/15 (the leaky protocol); Protocol B partitions by subject 70/15/15 without applying the leakage-graph component, series, or near-duplicate constraints; and Protocol C uses the frozen SPLITGUARD-AD component-safe split. Each protocol is run for five seeds, 15 epochs each, with AdamW at $\text{lr} = 10^{-4}$ and a class-balanced BCE-with-logits objective.

Under Protocol A, the random image-level split places nearly every test subject into the training partition: the mean test-subject contamination across the five seeds is 99.7%, with several seeds reaching 100%. Protocols B and C, by construction, achieve 0% test-subject overlap.

Table 9 reports the per-protocol test AUROC with paired-seed bootstrap intervals ($B=10,000$ resamples of the five-seed array, descriptive; see “Statistical conventions” at the start of §3). The total inflation gap (Protocol A – Protocol C) is +0.118 [+0.096, +0.145] AUROC, and the leaky->-SPLITGUARD-AD direction holds in all five seeds. A two-level hierarchical (subject $\times$ seed) re-analysis, the more defensible inferential quantity, gives +0.118 with an interval of [+0.061, +0.181] (see Robustness analyses below). The gap decomposes into a large subject-identity component (Protocol A – Protocol B, 0.110) and a small marginal beyond patient-level splitting (Protocol B – Protocol C, 0.008 [-0.033, +0.042]) whose interval touches zero.

Table 9: ADNI inflation gap, 5 seeds, 15 epochs, ResNet-18, coronal centre slice. Numbers are mean AUROC with paired-seed bootstrap 95% CI (B=10,000). “Test contam.” = mean % of test subjects also appearing in the train partition. The lower panel drops the 28 magnetization-transfer T1 scans for a sequence-homogeneity sensitivity check.

Protocol	AUROC [95% CI]	n_{test}	Test contam.
<i>All scans (MT1 included):</i>			
Random (leaky)	0.947 [0.926, 0.965]	169	99.7%
Subject-only	0.837 [0.810, 0.862]	169	0.0%
Component-safe (SPLITGUARD)	0.829 [0.813, 0.846]	165	0.0%
Total gap (random – component-safe)	0.118 [0.096, 0.145]		
Subject-leakage (random – subject-only)	0.110 [0.086, 0.139]		
Component-marginal (subject-only – comp.-safe)	0.008 [-0.033, 0.042]		
<i>MT1 acquisitions excluded:</i>			
Random (leaky)	0.960 [0.947, 0.973]	168	99.7%
Subject-only	0.833 [0.805, 0.857]	168	0.0%
Component-safe (SPLITGUARD)	0.831 [0.810, 0.855]	163	0.0%
Total gap (random – component-safe)	0.128 [0.097, 0.159]		
Subject-leakage (random – subject-only)	0.127 [0.103, 0.154]		
Component-marginal (subject-only – comp.-safe)	0.002 [-0.017, 0.028]		

Subject-level vs image-level evaluation (recommended clinical reading)

An image-level AUROC weights participants by scan count, and participants do not contribute equal numbers of scans. We therefore aggregate each participant’s per-scan predictions (mean y_{prob}) and recompute AUROC at the participant level on the converter-inclusive arm (Table 10): the gap survives the aggregation, and per-protocol AUROC is marginally higher under it, consistent with within-participant noise reduction. The image-level number is retained as the cross-cohort-comparable headline; *but the subject-level number is the recommended reading for clinical interpretation*, because it is a stricter aggregation that rules out per-subject scan-count heterogeneity as a confound and is closer to the patient-level unit of clinical decision-making. Downstream users of SPLITGUARD-AD report cards should quote the subject-level gap in clinical-adjacent contexts and the image-level gap only in benchmark-comparison contexts.

Robustness analyses

Four pre-specified arms test whether the headline gap depends on a particular methodological choice, and a fifth row reports the post-hoc size-balanced control of §4.2 (Table 11). All four preserve the leaky->-SPLITGUARD-AD ordering in every paired-seed bootstrap resample, and the total gap moves by at most 0.028 AUROC across them. The component marginal does not behave that way. It is Protocol B – Protocol C throughout, and it inverts to -0.026 on the converter-inclusive arm, opposite to the direction a leakage-removal effect would take, since there the component-safe constraints raised AUROC relative to subject-only splitting rather than lowering it. No interval excludes zero in any arm. We therefore report the total gap as robust and the component marginal as unresolved at five seeds, and we do not

Table 10: ADNI image-level against subject-level AUROC on the converter-inclusive arm (five seeds). Both columns are AUROC with a paired-seed bootstrap interval at $B = 10,000$. Subject-level numbers average each participant’s per-image predictions before computing AUROC.

Protocol	Image-level	Subject-level
Random (leaky)	0.957 [0.946, 0.969]	0.966 [0.962, 0.969]
Subject-only	0.789 [0.730, 0.851]	0.818 [0.771, 0.873]
Component-safe (SPLITGUARD-AD)	0.815 [0.795, 0.835]	0.829 [0.794, 0.858]
<i>Inflation gap (Protocol A – Protocol C):</i>		
Image-level	+0.142 [+0.131, +0.158], positive in 5/5 seeds	
Subject-level	+0.137 [+0.108, +0.173], positive in 5/5 seeds	

resolve it by adding seeds: §3.7 shows the margin is near-zero on intact cohorts by construction, so the informative axis is provenance loss rather than seed count.

Table 11: ADNI robustness arms; the primary arm excludes converters. Columns are the total gap (Protocol A–C), which is the paper’s headline, the subject part (A–B) and the component marginal (B–C), which is its null result; all three are measured on the same runs. Intervals are paired-seed bootstrap at $B = 10,000$.

Arm	A – C	A – B	B – C
Primary	+0.118 [+0.096, +0.145]	+0.110 [+0.086, +0.139]	+0.008 [–0.033, +0.042]
Converter-inclusive	+0.142 [+0.131, +0.158]	+0.168 [+0.105, +0.232]	–0.026 [–0.100, +0.048]
MT1 excluded	+0.128 [+0.097, +0.159]	+0.127 [+0.103, +0.154]	+0.002 [–0.017, +0.028]
DenseNet-121	+0.146 [+0.134, +0.161]	+0.136 [+0.107, +0.165]	+0.011 [–0.026, +0.053]
Size-balanced	+0.136 [+0.098, +0.159]	+0.110 [+0.086, +0.139]	+0.026 [–0.040, +0.068]

Table 12: Test-partition composition per protocol on the ADNI converter-inclusive arm (five-seed means), the arm on which the subject-level aggregation is computed. The three protocols hold image count roughly constant but differ fourfold in how many distinct participants those images come from, because tightening the split rule packs more scans of fewer people into the test partition. Every protocol-to-protocol AUROC comparison in this paper is conditioned on this asymmetry, and we report it here rather than leaving it implicit.

Protocol	Test participants	Test images	Images per participant
A (random)	160.8	213.0	1.32
B (subject-only)	48.0	220.0	4.58
C (component-safe)	38.0	203.2	5.35

The DenseNet-121 arm additionally replicates on OASIS-1 (+0.048, §3.2), so the inflation is consistent across two convolutional backbones on both source cohorts.

A two-level subject $\times$ seed hierarchical resampling of the primary arm widens the total-gap interval to $[+0.061, +0.181]$. Calibration is far worse under both honest protocols than under the leaky one, and not ordered between them (Brier/ECE: Protocol A 0.089/0.079, Protocol B 0.247/0.253, Protocol C 0.196/0.188) [45, 46]: the leaky protocol is calibrated to its own contaminated test distribution, which is not a property worth having [47]. Test-set demographic composition is balanced by construction across the five frozen splits (mean $50\% \pm 6\%$ female, $49\% \pm 7\%$ aged ≥ 76 ; 220 participants, median age 76).

Whole-volume input

The volumetric arm asks whether the shortcut belongs to the 2D slice or to the participant. A 3D ResNet-18 on 128^3 volumes, trained on the same 220 participants under the same three protocols, puts the total gap at $+0.235 [+0.174, +0.298]$, positive in 5/5 seeds and $[+0.145, +0.335]$ under the subject $\times$ seed hierarchical resampling: about twice the $+0.118$ of the 2D primary arm. The gap widens from both ends, and only one of those ends is about leakage. The leaky level rises to 0.971, against 0.947 in 2D, which is what a memorable unit of one whole volume rather than one slice predicts. The honest level falls to 0.736, against 0.829 in 2D, and that is not a leakage effect: this network is trained from scratch on 722 volumes where the 2D arms fine-tune ImageNet features, so its honest performance is not a competitive volumetric baseline and we do not offer it as one. The transferable quantity is the gap and not the level, here as everywhere else in this paper. The decomposition reproduces the 2D pattern: the subject layer carries the inflation ($+0.268 [+0.163, +0.372]$) while the component layer adds nothing resolvable at five seeds ($-0.033 [-0.185, +0.122]$, spanning zero, with subject-only at 0.703 falling below component-safe just as it does in the converter-inclusive 2D arm).

Site $\times$ label confound check

The ADNI CN/AD universe spans 48 recruiting sites (site code extracted from the first triplet of the ADNI PTID); on the full universe, uncorrected Cramér’s V between site and CN/AD label is 0.36 ($\chi^2 = 178$, $p < 0.001$), reflecting the well-known fact that ADNI sites differ systematically in their CN/AD subject mix. Within the five frozen component-safe SPLITGUARD-AD splits, this association is preserved rather than eliminated (script: `scripts/site_scanner_confound_audit.py`; JSON: `reports/tables/adni/adni_site_scanner_confound_audit.json`). Table 13 reports bias-corrected Cramér’s V (Bergsma’s small-sample correction [48]) for site $\times$ label and sex $\times$ label across the five primary-arm component-safe splits. The site-label association is moderate in train ($V \approx 0.36$), substantial in validation ($V \approx 0.48$), and strong in test ($V \approx 0.54$, ranging to 0.72 across seeds); the sex-label association is negligible in train and test but reaches $V \approx 0.34$ in the small validation partition. Scanner field strength is constant across this cohort and so cannot confound the label. The framework does not perform site-aware partitioning by design and is therefore not a substitute for site harmonisation (ComBat [34–36, 49]) or for explicit site-stratified splitting; the inflation gap we report is the subject-identity

channel that harmonisation does not fix post-hoc (§2.2). A site-stratified extension of the component-safe construction is named as future work in §4.2.

Table 13: Site and sex confound audit on the five ADNI primary-arm component-safe splits (1,125 manifest rows, 1,123 diagnosis-resolved; CN/AD rows only). Bias-corrected Cramér’s V (Bergsma [48]) between each covariate and the binary CN/AD label per partition; mean and range across the five seeds. The site–label association is substantial in every partition and grows as the partition shrinks; it is *exposed* by the audit but not *resolved* by the component-safe split, which is the central caveat of §4.2. The sex–label association is negligible in train and test but reaches $V = 0.34$ in the small validation partition, an instability we attribute to its size ($n = 165$) rather than to a population effect. Scanner field strength is constant (1.5 T) across every partition of this cohort, so $V(\text{scanner}|\text{label}) = 0$ by construction and the covariate is not tabulated.

Partition	Rows (mean)	Unique sites (range)	$V(\text{site} \text{label})$ (mean [min–max])	$V(\text{sex} \text{label})$
Train	793	38–39	0.360 [0.347–0.375]	0.000
Val	165	8–10	0.479 [0.464–0.495]	0.343
Test	165	11–12	0.538 [0.418–0.715]	0.046

Scanner field strength is constant (1.5 T) in every partition; $V(\text{scanner}|\text{label}) = 0$ throughout.

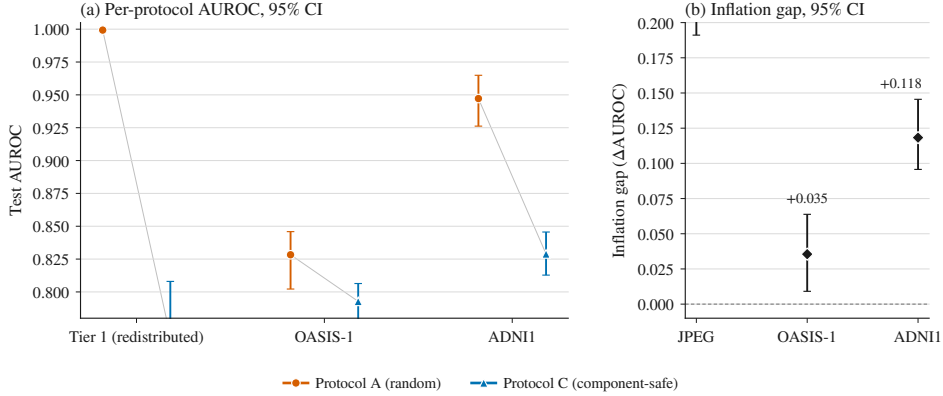


Fig. 6: Direction-preserved inflation gap on the two source cohorts and on the redistributed benchmark (ResNet-18). (a) Per-protocol test AUROC (random versus participant-safe) for each cohort; error bars are paired-seed bootstrap intervals ($B = 10,000$, $n = 5$ seeds per cohort; descriptive of seed-level variability, see “Statistical conventions” in §3). Grey lines connect the two protocols within each cohort. (b) The inflation gap ΔAUROC with the same descriptive intervals. On Tier 1 the participant-safe protocol is the identity recovered in §2.3.1, not the release’s own identifier, which is why its gap is larger here than the one a reader of that benchmark could measure. DenseNet-121 replications are reported in §3.1, §3.2 and §3.3; the figure shows ResNet-18 for visual parity with the rest of the paper.

Figure 6 places the ADNI gap in context under one reporting convention, paired-seed bootstrap intervals at $B = 10,000$. The gap is positive in the mean on both source cohorts: the small cross-sectional one (OASIS-1, $+0.035$ [$+0.009$, $+0.064$]) and the longitudinal one with substantial repeat structure (ADNI, $+0.118$ [$+0.096$, $+0.145$]). It is positive again, at $+0.233$ [$+0.191$, $+0.270$] (hierarchical [$+0.159$, $+0.311$]), on the redistribution of the first, where each participant contributes 32 slices rather than five. The three differ in how firmly, and we state the difference rather than averaging over it. On ADNI and Tier 1 the sign holds in every seed and both the paired-seed and the hierarchical intervals exclude zero. On OASIS-1 it does not: one of five seeds is negative (-0.010), and while the paired-seed interval excludes zero the subject $\times$ seed hierarchical interval is $[-0.033, +0.107]$, with the sign preserved in 84% of resamples. OASIS-1 is therefore a directional replication at the level of the seed mean, not an independently significant one, and we read it that way throughout. Patient identity is the dominant leakage channel on the two cohorts where a subject-only protocol exists to decompose it, ADNI and Tier 1; OASIS-1 carries no such arm. Subject to that, the random-split benchmark overstates AD-classification performance by a magnitude robust to the seed array, the converter-inclusion rule, the acquisition type, the CNN backbone and input dimensionality.

Figure 7 consolidates the headline inflation results across both source cohorts, the redistributed benchmark, both backbones and the ADNI robustness arms in a single reporting frame.

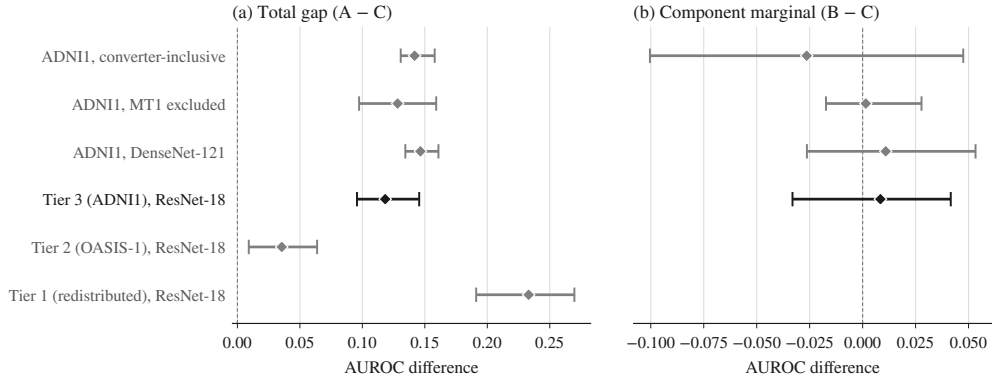


Fig. 7: Every inflation-gap arm, as point estimate and paired-seed bootstrap interval ($B = 10,000$, $n = 5$ seeds; descriptive of seed-level variability per the convention in §3). The primary arm (Tier 3, ResNet-18) is drawn in black. **(a)** Total gap, $\Delta\text{AUROC} = \text{Protocol A} - \text{Protocol C}$. The per-protocol AUROCs behind each arm are given in the corresponding subsection; the DenseNet-121 arms reproduce the gap rather than the absolute level ($+0.048$ on Tier 2 and $+0.146$ on Tier 3). **(b)** Component-layer marginal, Protocol B – Protocol C, on its own axis because it is an order of magnitude smaller. No interval excludes zero and the converter-inclusive arm inverts; the cross-cohort arms report no marginal and leave the slot empty. The two panels are the paper’s headline and its null result on the same runs. The ADNI hierarchical (subject $\times$ seed) bootstrap, the more defensible inferential cross-check, gives gap $+0.118$, CI $[+0.061, +0.181]$, direction preserved.

3.4 Is patient identity decodable, and does that explain the gap?

A linear probe (LinearSVC) trained on frozen penultimate-layer features (512-dim) of the 15 ADNI checkpoints of the converter-inclusive arm recovers participant identity far above chance, and to the same degree whichever protocol trained the representation: $211.7\times$ chance under component-safe splitting. Holding out a whole visit per participant instead of a random fifth of their images, so that no repeat acquisition from the same session sits on both sides, leaves identity decodable at $204.8\times$ chance. We run the probe on ADNI rather than on Tier 1 because ADNI supplies the participant identifiers the probe classifies; on Tier 1 the release’s key is not participant identity (§2.3.1).

The interpretation is narrower than the number invites. Identity is decodable from every one of these representations, including those trained under honest protocols, so the probe cannot explain a *difference* between protocols: a quantity that is the same in both conditions cannot account for the AUROC gap between them. What it establishes is a structural property of 2D MRI. Repeated slices of one head share shape, intensity profile and slice position, so identity is available to any visual representation, and that availability is what makes test-set contamination exploitable. This also qualifies the reading that saliency evidence invites: Rumala [28] observes

that a leaky network attends to subject-identifying structure and takes that as the mechanism of the inflation, but identity in the representation is a precondition for the shortcut rather than evidence that it was taken. The probe uses one architecture and one probe family; a non-linear adversary in the manner of DeepBrainPrint [16] or BrainPrint [15] would upper-bound it. The causal weight therefore rests on the two controlled interventions of §3.6 and §3.7.

3.4.1 Three-point degradation curve

The probe answers *whether* identity is encoded; the degradation curve answers *how much* of the inflation each grouping removes (Figure 8).

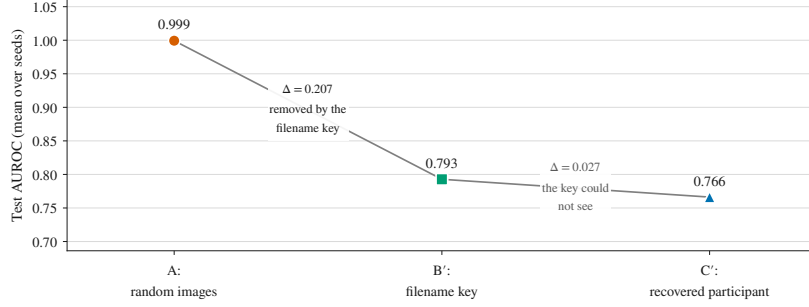


Fig. 8: Three-point degradation on Tier 1. Each step changes only what the split groups by: random images, then the filename key the release ships, then the participant identity recovered from OASIS-1. The first step is what an audit run inside the release can achieve; the second is what its lost provenance hides.

The $A \rightarrow B'$ step is $\Delta \text{AUROC} = +0.206$ and the $B' \rightarrow C'$ step a further $+0.027$. On ADNI, where identifiers are intact, the same decomposition puts $\sim 93\%$ of the gap in the first step. The two cohorts bracket the claim: grouping on a correct identifier removes most of the inflation, and grouping on a damaged one does not.

Decomposition takeaway (and head-to-head against GroupKFold)

The three-point degradation isolates $\sim 93\%$ of the ADNI inflation gap to subject-identity transfer; on Tier 1 the release’s own identifier recovers 88.6% of the gap, the remainder being the participants that identifier fails to group (§3.1). Protocol B enforces the same subject-independence constraint as GroupKFold, under a fixed three-way split rather than K -fold cross-validation (§1); we did not run GroupKFold itself, so this decomposition quantifies the marginal value of the component-safe layer above subject-grouped partitioning rather than constituting a head-to-head tool benchmark. That marginal, reported throughout as Protocol B – Protocol C, is 0.008 $[-0.033, +0.042]$ on the ADNI primary arm and -0.026 on the converter-inclusive arm (Table 11): small, of inconsistent sign, and with no interval excluding zero. On Tier 1 the comparison does not exist in that form, because the release carries no identifier that Protocol B could group on.

That marginal should be read with a further caveat that we did not appreciate until auditing our own splits. Protocols B and C do not differ only in their grouping. Protocol B shuffles participants and cuts at 70/15/15 by participant count; Protocol C consumes a frozen manifest produced by a size-descending greedy bin-packer that hits 70/15/15 by image count. The two therefore differ in the grouping (the variable of interest) and also in the assignment algorithm and the quantity the ratios are computed over. On a difference that small, the assignment step is not obviously negligible relative to the grouping step. The same control failure, measured directly in §3.7, was worth about half the apparent effect there. We therefore report the component marginal as an upper bound on what the grouping contributes rather than as an isolated estimate of it, which reinforces rather than weakens the conclusion already stated: the marginal is small, of inconsistent sign, and not resolved.

The greedy also leaves a visible fingerprint on the frozen ADNI manifests. Because it fills the validation and test bins first and takes the largest components that still fit, the partitions end up nearly disjoint in component size: across the five seeds, training components span 4–6 scans (mean 4.65), validation 7–9 (mean 7.50) and test 5–7 (mean 6.11). Participants with longer longitudinal records are systematically routed away from training. Class balance survives this, because components are stratified by label before the bin-packing runs (39.1%, 38.2% and 40.0% AD by image across the three partitions), but within each class the test partition is enriched for participants with more follow-up. That matters clinically: in this cohort AD participants contribute fewer scans than CN participants (4.55 ± 0.97 against 5.56 ± 1.00), so scan count tracks retention, and retention tracks disease course. This is an instance of the duration-dependent selection bias described in the data-handling literature [50], arriving through the splitter rather than through the cohort. We flag it here, report the distribution rather than leaving it implicit, and note in §4.2 what it means for the audit gate that this split passed every check the gate performs.

The dominant practical recommendation is therefore straightforward and not novel, split by subject identity. The leakage-graph machinery (component-level and near-duplicate edges) costs essentially nothing on cohorts with clean participant identifiers, and buys nothing measurable there either: we claim no *AUROC* benefit from it on ADNI or OASIS-1. §3.7 shows what it does buy, on the axis where it can: structural protection against patient leakage once subject identifiers start disappearing. We can, however, exhibit the failure mode it exists for, which the next paragraph does. We retain the component-safe construction as the conservative default in the pre-training GO/NO-GO gate; readers running the framework on cohorts where subject identity is clean and converter-free should expect the component-safe and subject-only protocols to be statistically indistinguishable. The honest contribution claim is therefore infrastructural (an executable AD-specific audit with a binary gate), not that the component-graph layer outperforms a careful subject-aware GroupKFold on every cohort.

Where grouping by the available identifier is blind

Subject-wise splitting can only protect images whose participant is known, and a redistributed benchmark is exactly the case where that premise fails without saying

so. Tier 1 makes the failure measurable, because its true identities are recoverable from outside the release (§2.3.1), which is not something its users have.

The exposure is not a handful of edge cases but the structure of the release, measured in §2.3.1: a key that merges and splits participants, images with no key at all, and a split that every check the release supports calls clean while most of its test set is contaminated. This is the regime the component layer exists for, and the honest reading of Tier 1 is that the layer does not rescue it either, because the redistribution removed the session and series identifiers the graph would have carried. The consequence for a reader of public benchmarks is stronger than a recommendation to split by subject: where a release does not document provenance, a subject-wise split of it cannot be verified from inside the release at all, and the number it reports cannot be interpreted.

3.5 What the gap costs at a screening operating point

Running both leaky and honest protocols on the same ADNI universe is not just a robustness check; it surfaces *two* findings that the leaky benchmark would have concealed. First, a sex-related performance disparity emerges under both honest protocols and collapses to indistinguishable from zero under the leaky one (Supplementary Material S6). Sex differences in AD are well documented: in atrophy rates on ADNI itself [51], in verbal-memory reserve that can delay diagnosis in women [52], and in incidence and prevalence [53], so a protocol that conceals a performance disparity between sexes conceals something the clinical literature says should be looked for. We report ours as exploratory and do not attribute it to any of these mechanisms. Second, the AUROC gap translates into a concrete clinical cost-of-leakage at the screening operating point that a clinician trusting the leaky number would systematically underestimate (§3.5.1). Both findings rely on the per-image test predictions persisted by the converter-inclusion sensitivity arm; both rely on the framework’s audit-and-prevent design to render them visible.

3.5.1 Cost-of-leakage as an arithmetic illustration

An inflation gap of +0.118 test AUROC is concrete only to readers who think in AUROC. To re-express the gap in units a clinician can weigh, we compute the sensitivity each protocol achieves at a fixed screening specificity and then convert the difference into missed-diagnosis counts per 1,000 screened patients via the deterministic identity $\text{missed}/1000 = 1,000 \cdot \text{prev} \cdot (1 - \text{sens})$ at literature-cited reference prevalences. This is an *arithmetic translation* of the AUROC gap into prevalence-weighted units, not a prospective deployment claim: the underlying sensitivities come from the ADNI test partitions and inherit all cohort caveats listed in §4.2 (single 2D coronal-centre slice, no external clinical validation, no real-world workflow integration).

Operating point

For each (seed $\times$ protocol) on the converter-inclusive ADNI sensitivity arm we extract the held-out ROC curve from the persisted per-image test predictions and read off

the sensitivity at fixed specificity = 0.90, a conventional triage-assistant anchor that bounds the acceptable false-positive workload at 10% of screened negatives. We choose this anchor rather than a stricter (≥ 0.95) population-screening threshold or a more permissive (~ 0.80) memory-clinic triage threshold because it is the middle ground at which the leaky-vs-honest disagreement is most visible in absolute counts; the qualitative direction of the cost-of-leakage gap holds at any specificity above the no-discrimination diagonal. Averaged across the five seeds, sensitivity at specificity = 0.90 is 0.866 ± 0.046 under Protocol A (leaky), 0.551 ± 0.105 under Protocol B (subject-only), and 0.535 ± 0.103 under Protocol C (SPLITGUARD-AD). Youden’s J preserves the leaky-versus-honest separation ($J = 0.785/0.458/0.533$ for Protocols A / B / C; $A - C = +0.252$) but *inverts* the B–C order relative to the fixed-specificity reading: under J the component-safe protocol scores above subject-only. That inversion is the same one seen in the converter-inclusive component marginal (§3.3), and we flag it rather than average over it: the B–C contrast is not stable to the choice of operating point, which is itself evidence that the component layer’s marginal is within noise on this cohort. Figure 9(a) shows the corresponding ROC curves with the operating point marked.

Threshold selection

Every operating point in this section is *selected on the validation partition and measured on test*. This distinction is not cosmetic: scanning the test ROC for the threshold that maximises sensitivity at a target specificity (the natural implementation, and the one we used initially) reports the best value obtainable in hindsight rather than one obtainable prospectively, and would make this manuscript commit a version of the error it exists to measure. Choosing thresholds honestly lowers the reported sensitivity of *both* protocols, but it lowers the component-safe protocol further (its sensitivity at spec = 0.90 falls from 0.594 under test selection to 0.533 under validation selection, against 0.874 to 0.865 for the leaky protocol), because the leaky model’s test performance was already inflated and had less to gain from a retrospectively chosen cut. The corrected leaky-vs-honest gaps are therefore *larger*, not smaller.

Robustness to operating-point choice

The leaky-vs-honest sensitivity gap is uniformly positive across every operating point checked and monotone in the specificity anchor (+0.270 at spec = 0.80; +0.331 at spec = 0.90; +0.424 at spec = 0.95; Youden-J-optimal +0.153 in sensitivity and +0.252 in J). The 0.90 specificity anchor used for the cost translation is therefore a middle pick: the gap is smaller at looser specificities and larger at 0.95, so the anchor neither maximises nor minimises the reported cost. The full ten-row operating-point table is provided as Supplementary Table S2.

Reference prevalences

We anchor the translation on two literature-cited prevalence values matched to plausible deployment scenarios for an ADNI-trained CN→AD classifier: (i) 13.8% (95% CI 13.1–14.5%) population AD prevalence in US adults 75–84 in 2020 [54], the natural anchor for ADNI’s median age of 76, and (ii) 58.9% MCI-or-dementia diagnostic

yield at a tertiary memory clinic [55], anchoring the high-prevalence end where the model would act as triage downstream of a referral filter. The clinical-cost framing is established by [56–58]. Because ADNI is a research-recruited volunteer cohort with enrichment for memory complaints, the $\sim 3.5\times$ underestimate is best read as a calibration of the AUROC gap’s clinical-cost *magnitude* rather than as a prospective deployment estimate.

Headline arithmetic translation

At the screening operating point (specificity = 0.90), the arithmetic difference between the leaky benchmark’s apparent sensitivity (0.866) and Protocol C’s honest sensitivity (0.535) corresponds to the missed-diagnosis shortfall summarised in Figure 9(b). At the population-screening prevalence of 13.8% the leaky benchmark’s sensitivity arithmetically implies ≈ 18.5 missed AD diagnoses per 1,000 screened older adults; the honest sensitivity implies ≈ 64.2 , a difference of 45.7 missed cases per 1,000, equivalently, a $\sim 3.5\times$ arithmetic underestimate of missed-diagnosis volume. At the tertiary-clinic anchor of 58.9%, the same arithmetic underestimate amplifies to ≈ 195.0 additional missed cases per 1,000 (≈ 79.0 apparent against ≈ 274.0 actual), because the same sensitivity shortfall is applied to a larger diseased fraction.

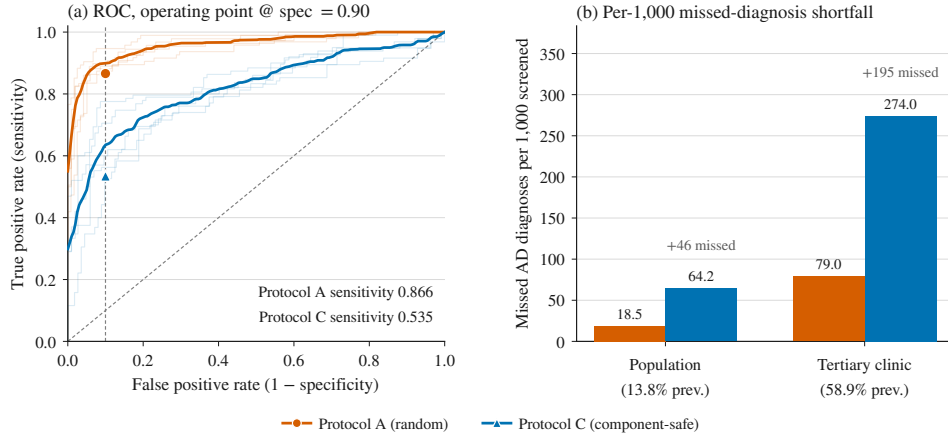


Fig. 9: Arithmetic cost-of-leakage translation on the converter-inclusive ADNI sensitivity arm. (a) Per-seed ROC curves for Protocol A (leaky) and Protocol C (component-safe SPLITGUARD-AD); the dashed vertical line marks the screening operating point at specificity = 0.90 (i.e. false-positive rate = 0.10); markers show the per-seed mean sensitivity at that specificity. (b) Missed AD diagnoses per 1,000 screened patients at the two literature-cited prevalence anchors, computed as $1,000 \cdot \text{prev} \cdot (1 - \text{sens})$. Protocol A’s bars use its apparent sensitivity (0.866) and Protocol C’s bars its honest sensitivity (0.535). The annotation “+N missed” is the per-1,000 arithmetic shortfall implied by reporting the leaky number instead of the honest one;

Scope

The translation is an arithmetic re-expression of the sensitivity gap under a fixed specificity anchor, not a prospective deployment estimate: the underlying sensitivities come from the ADNI test partition and inherit all cohort caveats in §4.2. The anchor is not load-bearing: Figure 10 sweeps it, and alternative operating points (Youden J, fixed-sensitivity) yield the same qualitative conclusion (Supplementary Table S2). A fully calibrated decision-curve analysis is a deployment-time tuning concern outside the scope of the leakage audit itself [45–47].

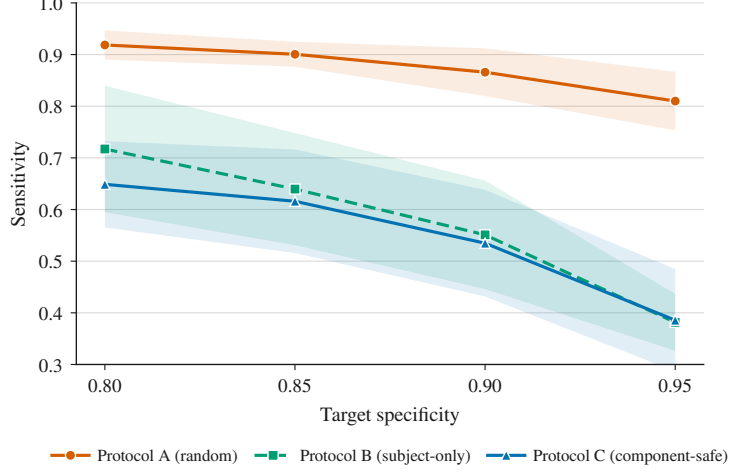


Fig. 10: Sensitivity against target specificity on ADNI1, mean over five seeds with ± 1 SD bands. The leaky protocol’s advantage is not an artefact of the 0.90 anchor: it holds across the sweep, and widens as the operating point tightens, which is the direction that matters for screening. Protocol B crosses below Protocol C near 0.95, the same inversion the component marginal shows in Figure 7(b); Protocol B’s band is the widest of the three, so the component layer’s apparent advantage over subject-only splitting is not stable to the choice of anchor.

3.6 Leakage as a dose: injecting test-set overlap

The cross-tier inflation gap and the cost-of-leakage translation both take the leakage condition as binary (leaky vs. honest). A more informative question for benchmark designers is the dose-response: if $p\%$ of test subjects also appear in train, how much AUROC inflation should we expect? We answer this with a controlled stress-test on the converter-inclusive ADNI sensitivity arm.

Design

Starting from the Protocol C component-safe split, we generate a family of injected splits at target contamination fractions $p \in \{0, 0.25, 0.50, 0.75, 1.0\}$, where p is the

share of test scans that belong to a training participant, not the share of test participants. For each non-zero p we draw $k = \lfloor p \cdot 258 \rfloor$ participants from the multi-visit training subset and substitute one alternate scan of each into the test set, displacing an equal number of original test scans. Because each donor contributes a single scan, the share of contaminated test *participants* is always the larger number, which is why every composition below is done in scan units. Class balance of the *injected partition* (118 CN / 140 AD over 258 rows) is held fixed at every overlap level by stratified substitution. The partition the model is *evaluated* on is not held fixed, and this qualifies the calibration: the binary pipeline drops MCI scans, substitution displaces them, and the evaluated CN/AD subset therefore grows from 207 rows at $p = 0$ to 258 at $p = 1$, with the AD share rising from 46% to 54%. The dose axis is thus partly confounded with test composition, so the slope should be read as the response to contamination *and* the modest enrichment that accompanies it, not to contamination alone. We report it rather than correct it because the two cannot be separated without changing the injection mechanism, and because the direction of the confound inflates nothing the paper relies on: the slope is used only as a floor (§3.5.1). Otherwise this is biometric-identity leakage alone: no slice or near-duplicate is shared between train and test, only the patient. We run the full 5 seeds $\times$ 5 overlaps matrix on two architecture families (ResNet-18 and DenseNet-121, 15 epochs each, $\text{lr} = 10^{-4}$ AdamW), a total of 50 training runs.

Result

Figure 11(a) shows the resulting per-seed AUROCs and the seed-averaged dose-response curves with 95% paired-bootstrap CIs. The seed-averaged curves increase with test-subject overlap for both architectures; per-seed monotonicity holds in 5/5 ResNet-18 seeds but in 0/5 DenseNet-121 seeds (Supplementary S4); the fitted ResNet-18 endpoints (0.823 at $p = 0$, 0.948 at $p = 1$) sit within 0.02 of the converter-inclusive arm’s honest and leaky levels (0.815 and 0.957, §3.3), which is the arm the injected splits are built from. We read that agreement as a sanity check and not as a calibration: the matrix and those levels come from the same code state on different GPUs (§Declarations), and nothing in the injection was tuned to reproduce them. The straight-line OLS fit (Figure 11(b)) recovers

$$\widehat{\text{AUROC}}_{\text{ResNet-18}} = 0.823 + 0.125 \cdot p, \quad R^2 = 0.853, \quad (1)$$

and a similar but noisier curve for DenseNet-121 ($\widehat{\text{AUROC}} = 0.861 + 0.091 \cdot p$, $R^2 = 0.64$). The slope of +0.125 AUROC per unit overlap on ResNet-18 implies a cohort- and architecture-specific calibration point: on this ADNI universe and this 2D-slice representation, every 10 percentage points of test-subject overlap inflates the reported AUROC by approximately +0.013. The point-estimate predictions from Equation 1 are usefully read as the deterministic part of a noisy calibration; the per-seed scatter around the regression line in Figure 11(b) is approximately consistent with an additive Gaussian residual, so the dose-response can equivalently be written

$$\text{AUROC}(p) = \alpha + \beta \cdot p + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad (2)$$

with $(\hat{\alpha}, \hat{\beta}) = (0.823, 0.125)$ and $\hat{\sigma} = 0.0191$ on ResNet-18 (0.0250 on DenseNet-121), the residual standard deviation of the twenty-five per-seed observations about the OLS fit. The unit matters: because the band is meant to describe a *single* training run, it has to be estimated from single runs. Taking it instead from the five per-overlap means would give 0.008 and understate the spread roughly 2.5-fold, since averaging five seeds removes most of the seed-level variance the band exists to represent. So estimated, $\hat{\sigma}$ gives an order-of-magnitude prediction band for one run at a given p and lets the equation be used probabilistically (*e.g.*, the probability that a benchmark reporting AUROC A at measured contamination p exceeds its true component-safe value by more than 0.05 can be estimated from the upper tail of $\mathcal{N}(A - \hat{\alpha} - \hat{\beta}p, \hat{\sigma}^2)$). We report the OLS slope as a descriptive calibration constant rather than as an inferential coefficient (the twenty-five (seed, overlap) observations are not independent under OLS). A random-intercept model fitted by iterated feasible GLS under compound symmetry gives the same slope, $\hat{\beta} = 0.1251$, because the design is balanced; what changes is the interval. With a cluster-robust sandwich standard error carrying the finite-cluster correction and a $t(4)$ critical value, the 95% CI is $[0.0866, 0.1636]$ on ResNet-18; DenseNet-121 gives $\hat{\beta} = 0.0906$ with 95% CI $[0.0416, 0.1396]$. Both exclude zero. Full specification, the fitted variance ratio, the seed-cluster bootstrap, and Spearman monotonicity statistics are in Supplementary Material S4. Density-choice qualifications and finer-grid trade-offs are discussed in §4.2.

Interpretation

The dose-response is approximately linear over the entire $[0, 1]$ overlap range, with one qualification we state rather than smooth over: the ResNet-18 increments shrink monotonically across the sweep (+0.046, +0.035, +0.025, +0.020) and the DenseNet-121 mean falls slightly at the last step (0.9403 at $p = 0.75$ to 0.9387 at $p = 1$). That is a mild saturation as the leaky endpoint is approached, which is what a bounded metric does near its ceiling, and not a threshold below which contamination is harmless. No threshold effect appears anywhere in the range, which is consistent with the biometric-probe finding (§3.4) that subject identity is encoded as a roughly continuous signal in 2D MRI representations. Both architectures' $p = 1$ endpoints converge near the Protocol A leaky AUROC, as expected: 100% subject overlap is mechanically equivalent to the random-split condition. The DenseNet-121 curve has a higher baseline AUROC at $p = 0$ (0.853 vs. 0.814) and a flatter slope (+0.091 vs. +0.125). The two mixed-effects slope intervals overlap ($[0.0866, 0.1636]$ vs. $[0.0416, 0.1396]$; Supplementary S4) and we run no formal difference test, so the two architectures' dose-response slopes are not distinguishable at five seeds and we draw no inference about backbone capacity. The DenseNet R^2 of 0.64 still trails the ResNet-18 fit, and no DenseNet seed is monotonic across all five overlaps: with five seeds per overlap and a smaller endpoint separation, the trend is carried by the seed average rather than visible within any single seed.

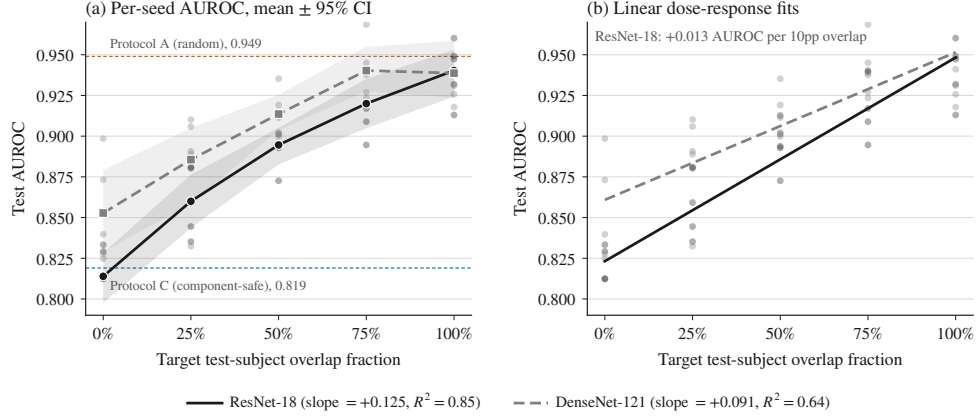


Fig. 11: ADNI leakage dose-response on the converter-inclusive sensitivity arm. (a) Per-seed test AUROCs (small dots), seed-averaged mean (markers + line) and 95% paired-bootstrap CI band for each architecture as a function of the target test-subject overlap fraction p ; ResNet-18 is drawn in black and DenseNet-121 in grey (dashed), and the reference lines take the colour of the protocol they mark. Reference lines mark the *primary-arm* Protocol C component-safe baseline (0.829) and Protocol A leaky benchmark (0.947) from §3.3, shown for cross-arm comparability. The matrix itself is built on the converter-inclusive arm, whose own levels are 0.815 and 0.957; the fitted endpoints are 0.823 and 0.948 (panel a). (b) OLS linear fits to the per-seed points show the dose-response slope; the ResNet-18 slope of +0.125 AUROC per unit overlap is a cohort- and architecture-specific calibration point on this ADNI universe and this 2D-slice representation, not a claim that the slope generalises to other cohorts or 3D representations.

Practical use, and why the slope is a floor

Equation 1 converts a measured test-subject contamination fraction (which the pre-training audit already reports) into expected AUROC inflation. It should be read as a lower bound on the gap rather than as an estimate of it, for a reason that follows from how it was fitted: the injection varies test-subject overlap with the training set held fixed, so it prices the participant-identity channel and no other. A naive split on a real cohort additionally leaks through slices of one volume, near-duplicates and re-acquisitions, none of which the slope sees.

The two cohorts where we measured the total gap bear this out. Under random splitting, contamination is essentially complete on both: on ADNI1 because participants contribute about five scans each, and on Tier 1 because a participant’s 32 slices are all but certain to land on both sides of the split. The equation therefore predicts a drop of at most 0.125 on each, against a measured +0.142 on the converter-inclusive ADNI1 arm the injected splits are built from and +0.233 on Tier 1, and the larger shortfall on Tier 1 is what one would expect where adjacent slices of one volume add a channel the slope does not price. On the converter-excluding primary arm the measured gap is +0.118, which the predicted floor now matches to within 0.01: the margin

the floor leaves is a property of the cohort it is applied to, not a guarantee, and on a cohort whose identity channel is the whole of its leakage the floor is the estimate. Used as a point estimate the equation would tell a reader their honest AUROC is higher than it is, the wrong direction to err in for a quantity whose purpose is to correct over-reporting. Used as a floor it answers the question worth asking of a benchmark: whether the contamination present is large enough to matter at all.

3.7 Provenance loss: what the leakage graph buys when identifiers fail

§2.2.2 reports that on every cohort studied here the connected components coincide exactly with a subject grouping, so the component layer earns nothing over GroupKFold-style splitting. That is a consequence of subject being the coarsest identifier: session, series and acquisition-date edges all nest inside it, so none of them can change a component while `subject_id` is present and correct. The framework’s stated scope condition is therefore that the extra edge families matter only where subject identity is absent, corrupted, or inferred wrongly. Until now that was an assertion. This section measures it.

Design

The regime is created deliberately, using the same template as the dose-response experiment in §3.6: hold everything fixed, sweep one controlled quantity, read off the curve. Here the swept quantity is provenance loss. Starting from the ADNI1 primary arm, where subject identifiers are complete and authoritative, we delete `subject_id` on a random 0/10/25/50/100% of scans and ask what each protocol does with the remainder. Deletion is applied per scan rather than per subject because that is the observed failure mode: redistributed benchmarks lose the patient index one filename at a time, so a record is typically partly recoverable rather than wholly anonymous. Under Protocol B every scan whose identifier was removed becomes its own group and may be placed in any partition. Under Protocol C the leakage graph is rebuilt from whatever identifiers survive; `session_id` and `series_uid` are globally unique on this cohort (1,003 and 1,008 distinct values, no blanks, and no value shared between two participants) and do not take `subject_id` as an input, so they continue to bind scans that `same_subject` can no longer reach.

The outcome is scored against the untouched ground-truth identifiers, which the degradation preserves out of band. We report the number of true participants with scans on both sides of a partition boundary. This is a structural measurement: it counts the leakage each protocol actually admits without routing the question through a trained model, and it is therefore free of the seed-variance and test-set-size caveats that qualify the AUROC comparisons elsewhere in this paper.

Result

Figure 12 and Table 14 give the curve. At zero loss the two protocols admit no leakage and are indistinguishable, reproducing the equivalence reported in §2.2.2. They separate immediately thereafter: at 10% identifier loss the leakage graph prevents 49%

Table 14: Provenance degradation on the ADNI1 primary arm (220 participants, five seeds, mean $\pm$ SD). Columns report participants whose scans straddle a partition boundary, and p , the fraction of test-partition participants who also appear in training, the same quantity the dose-response experiment sweeps, so the two curves compose. At zero loss the protocols are identical, which is the §2.2.2 null result restated as the origin of a curve rather than as a dead end.

Identifiers deleted	Straddling participants		p ($\text{test} \cap \text{train}$)	
	Protocol B	Protocol C	Protocol B	Protocol C
0%	0.0 ± 0.0	0.0 ± 0.0	0.000	0.000
10%	18.6 ± 5.0	9.4 ± 1.9	0.361	0.207
25%	31.4 ± 3.1	19.8 ± 2.0	0.374	0.408
50%	77.0 ± 5.0	53.6 ± 6.6	0.980	0.807
100%	182.6 ± 7.2	125.4 ± 1.3	0.993	0.990

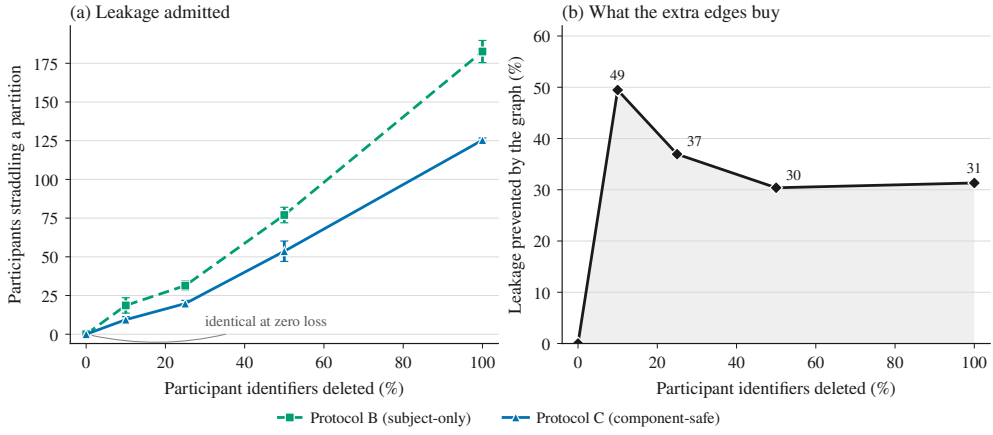


Fig. 12: Provenance degradation on the ADNI1 primary arm, five seeds, mean $\pm$ SD. **(a)** Participants whose scans straddle a partition boundary as participant identifiers are deleted. This counts admitted leakage directly, without routing the question through a trained model. The two protocols are indistinguishable at zero loss (the equivalence reported in §2.2.2, here the origin of a curve rather than a dead end) and separate immediately thereafter. **(b)** The share of that leakage the leakage graph prevents. Both protocols are assigned by the same class-stratified bin-packer, so the difference is attributable to the grouping and not to the assignment step; under a mismatched assigner the 10% figure reads 78% rather than 49%. The benefit peaks in the regime where redistributed benchmarks sit and decays once too few identifiers survive for any rule to bind on.

of the participant leakage subject-wise splitting admits (9.4 versus 18.6 straddling participants), and 31–37% at higher loss rates.

Both protocols are assigned by the same class-stratified bin-packer over their respective groupings, so the difference is attributable to the grouping and not to the assignment step. That control matters more than it may appear. An earlier version of this comparison assigned Protocol B by shuffling subject groups and cutting at 70/15/15 by group count, while Protocol C bin-packed by image count; under that mismatched design the graph appeared to prevent 78% of residual leakage at 10% loss. Roughly half of the apparent benefit came from the assigner rather than from the grouping. We report the matched comparison because only the grouping is the claim.

Provenance loss amplifies non-linearly

Deleting 10% of identifiers returns 36% of test participants to the training partition under Protocol B, a roughly threefold amplification. The mechanism is elementary once stated: a participant contributes a median of five scans, and one of them losing its index is enough to detach that scan from the group and let it land opposite the other four. Participant-level protection is not degraded gracefully by scan-level provenance loss; the first missing identifier in a record breaks it.

Composing the two curves, and where the composition fails

Chaining the two curves requires them to share a unit. The injection experiment sweeps the share of contaminated test *scans*, so that is the quantity we compose through: at 10% identifier loss it is 35% under Protocol B and 18% under Protocol C, against 36% of test *participants* under Protocol B. Read through Equation 1, that is +0.043 AUROC of expected optimism under Protocol B against +0.022 under Protocol C. The slope and the shares come from different arms of one code state (§Declarations), which is one more reason to read the composition as an order of magnitude rather than a coefficient. Table 14 reports the participant share p alongside it, because that is what a split audit shows a reader.

We do not present that as a general result, because p is not monotone across the sweep. At 25% loss Protocol C’s p (0.408) exceeds Protocol B’s (0.374), inverting an ordering that holds at every other level and that the straddling-participant count preserves throughout. The reason is that p is a ratio whose denominator is the number of distinct test participants, and the two protocols do not produce test partitions containing the same number of them: component-safe grouping yields larger and fewer components, hence a smaller and differently composed test partition. p is therefore a defensible input to the dose-response fit, where it is measured within a single protocol, but a poor instrument for comparing protocols. The straddling-participant count, which counts events rather than ratios, does not have that defect, and it is the measurement the result above rests on. We give the composition at 10% as an illustration of what chaining the two experiments would yield, and explicitly not as a validated protocol-level comparison.

Scope

Three limits belong with this result. The measurement is primarily structural: it counts admitted leakage, and the AUROC arm trained on the same degraded manifests is reported in §4.2 rather than here. The degradation is synthetic, imposed on a cohort with complete provenance rather than observed on one that lost it, which is what makes ground-truth scoring possible and also what makes it a model of the failure rather than the failure itself. And the recovery depends on which identifiers a cohort happens to carry: ADNI1 supplies globally unique session and series keys, whereas the Tier-1 benchmark supplies neither, so the curve measured here is an upper bound on what the same protocol could achieve on a cohort stripped down to filenames.

4 Discussion

Table 15: SPLITGUARD-AD benchmark report card for the ADNI1 primary-arm ResNet-18 model under the component-safe split (Protocol C), seed means over five seeds. The card illustrates the reporting format the framework produces; it is not a clinical-deployment artefact. PPV/NPV rows use the same prevalence anchors as §3.5.1.

SplitGuard-AD Benchmark Report Card	
Model Claim Boundary	CN versus AD from one 2D coronal-centre slice per scan, labels from the ADNI diagnosis table. Not intended for standalone clinical diagnosis.
Identifier provenance	ADNI PTID, session and series identifiers, all present for every scan.
Data Safety Audit (<i>blocking checks</i>)	
Participant Overlap	0% (✓Pass)
Session Overlap	0% (✓Pass)
Connected Component Overlap	0% (✓Pass)
Reported, non-blocking	
Label join	22% of labels by 180-day date proximity (§2.3.3)
Site-label association	Cramér’s $V = 0.54$ in test (Table 13)
Component-size balance	Warning: partitions differ in scans per participant (§3.4.1)
Debiased Performance Estimates (Held-out Test, $n \approx 165$ per seed)	
Sensitivity	65.5%
Specificity	81.0%
AUROC	0.829
Balanced Accuracy	73.2%
Prevalence-conditioned predictive values (illustrative)	
PPV @ 13.8% (population) [54]	35.6%
NPV @ 13.8% (population)	93.6%
PPV @ 58.9% (memory clinic) [55]	83.2%
NPV @ 58.9% (memory clinic)	62.1%
Computed from sensitivity/specificity above; predictive values are prevalence-dependent and tracked at the same two anchors as §3.5.1.	

To illustrate the reporting format that SPLITGUARD-AD produces alongside its splits, Table 15 gives a benchmark report card for the ADNI primary-arm model under the component-safe split. It synthesises the model’s claim boundary, where its identifiers come from, the blocking audit, the findings the audit reports without blocking, and the debiased performance estimates a reader would consult before treating the result as more than a benchmark score. The provenance row is the one Tier 1 motivates: the same three blocking checks pass on a Tier-1 manifest built from the release’s own identifier, and only the provenance row would have shown that nothing guaranteed they meant anything.

4.1 Clinical and field-level implications

At a fixed screening specificity of 0.90, ADNI sensitivity falls from 0.866 under random splitting to 0.535 under component-safe evaluation (§3.5.1): 45.7 additional missed AD cases per 1,000 screened at the US 75–84 population prevalence anchor. A model whose sensitivity moves by a third between its benchmark and its honest evaluation is not a standalone screening tool. This gap arises from a design choice, the evaluation protocol, that is invisible in most published papers, which is the precise reason SPLITGUARD-AD reports the audit as a binary pre-training GO/NO-GO gate rather than as a post-hoc diagnostic. Two clinical-scope notes follow. *First*, CN-vs-AD with MCI excluded is the diagnostically easier cell of the AD imaging task; the cell that matters most for early-intervention decisions, CN-vs-MCI or stable-vs-converter MCI, is also the cell where subject identity carries the most signal relative to the disease signal, so a leakage audit applied to those tasks would almost certainly surface a larger inflation gap than the one we report here. *Second*, the headline absolute AUROC numbers (0.829 honest, 0.947 leaky) are floor estimates from a single 2D coronal-centre slice and should not be read as performance ceilings; the transferable quantity is the *gap*, not the level.

Beyond the headline number, two scoping points matter for clinical readers. First, structural MRI is not a standalone diagnostic tool for AD; in clinical practice it informs the dementia workup by excluding structural mimics, by demonstrating atrophy patterns consistent with neurodegeneration, and by supporting clinical and biomarker-based diagnosis. A deep learning model evaluated on single slices labelled by a clinical rating cannot claim to diagnose Alzheimer’s disease; it can only claim to classify images by that rating in a controlled benchmark setting, and SPLITGUARD-AD makes this distinction explicit through its clinical claim boundary and label-confidence fields. Second, lower leakage-safe performance should not be read as a failed model; it is a more honest estimate of what remains after removing patient-identity and acquisition shortcuts.

For redistributed public datasets, near-perfect internal scores are not interpretable unless the release documents provenance that makes a participant-safe split checkable: Tier 1 shows that a manifest built on a release’s own identifier can pass every overlap check and still leave most of its test set contaminated. For metadata-rich cohorts such as OASIS-1, random slice splitting can still contaminate nearly every test subject, even when the resulting AUROC inflation is smaller and more seed-sensitive than on the redistributed benchmark. Future AD MRI studies should therefore release split

manifests, report subject overlap checks, document where both labels and identifiers come from, include calibration and sensitivity-at-fixed-specificity reporting, and validate on cohorts whose subjects, scanners, and preprocessing pipelines are independent of training.

4.2 Limitations

- The component layer is equivalent to subject grouping on intact cohorts, and validated beyond it only under synthetic provenance loss.** On the two source cohorts every connected component contains exactly one participant (Table 3): the session, series-UID and acquisition-date rules fire, on 223, 209 and 222 ADNI pairs, but each joins a pair `same_subject` had already joined. The marginal over Protocol B is accordingly small and unstable: 0.008 [-0.033, +0.042] on the primary arm, and ≤ 0 in three of four configurations, including a sign inversion on the converter-inclusive arm (Table 11). No interval excludes zero in any of them, and we claim no AUROC benefit on intact cohorts. §3.7 does establish a benefit, but on a regime we imposed rather than observed: it deletes identifiers uniformly at random from a cohort with complete provenance, whereas real redistributed benchmarks may lose them with structure our model does not reproduce: by site, by archive batch, or correlated with class. That measurement is also structural rather than predictive; it counts admitted leakage; its AUROC arm, trained on the same degraded manifests, gives 0.866 under subject-wise splitting against 0.829 under component-safe splitting at 10% loss. Tier 1 supplies the observed counterpart (§3.4.1), and on it the component layer does not recover what the identifier lost, because the redistribution removed the session and series identifiers along with the participant one. A cohort whose damage leaves partial provenance for the graph to carry is still the most informative next experiment. The near-duplicate detector is now calibrated against recovered identity (precision 0.986 and adjacent-slice recall 0.246 at $d \leq 4$); it ships report-only because no threshold we measured is both safe and useful (§2.2.2).
- Every finding is a property of a 2D coronal-centre-slice representation.** The inflation gap, the dose-response slope, the provenance curve, the subgroup disparity and the clinical translation are all measured on one slice per volume, chosen to keep ADNI and OASIS-1 comparable; OASIS-1 additionally derives five slices per processed volume, so its operating-point metrics are small-sample and seed-sensitive. ResNet-18 carries the primary results, with a pre-specified DenseNet-121 arm replicating the headline gap. Whether these quantities transfer to volumetric models is now partly answered rather than open: the volumetric arm (128³ volumes, 3D ResNet-18, `scripts/train_adni_3d.py`) runs against the same frozen partitions and puts the gap at +0.235 (§3.3), so the identity shortcut survives whole-volume input and is in fact larger there. What does not transfer is the *level*: that network trains from scratch on 722 volumes while the 2D arms fine-tune ImageNet features, so its honest AUROC of 0.736 is a weak volumetric baseline and the arm settles the direction of the effect, not the achievable performance of 3D models. A pretrained or larger-cohort volumetric replication remains open. Multiple-instance

aggregation over per-slice predictions and transformer backbones remain open, and reuse the same frozen manifests.

- **Label and identifier provenance differs by tier.** Tier-1 labels are CDR, but its participant identities are recovered by content matching rather than supplied, and we release the matching script against OASIS-1 rather than the mapping itself. Because all 200 Tier-1 participants are also in Tier 2, the two tiers are not independent replications of each other. On ADNI, 22% of labels (21% within the CN/AD subset) are assigned through a 180-day date-proximity join rather than an exact visit key (§2.3.3), and 28 magnetization-transfer T1 acquisitions (17 in the CN/AD universe) are non-MPRAGE; a pre-specified arm dropping them yields $+0.128$ [$+0.097$, $+0.159$] against $+0.118$ [$+0.096$, $+0.145$]. Restricting evaluation to the test scans whose label comes from an exact visit key gives a gap of $+0.113$; that tests the labels the gap is scored against, not the ones it was trained on, since the date-proximity scans remain in the training partition.
- **The intervals are descriptive, not inferential.** Five seeds support paired-seed bootstrap intervals that propagate between-seed variation only. The hierarchical subject $\times$ seed bootstrap, which also propagates participant-level sampling error and yields intervals $1.9\text{--}4.7\times$ wider depending on the arm, is run on every arm. The three protocols also do not place comparable numbers of patients in the test partition (Table 12): Protocol A averages 160.8 distinct participants per seed, Protocol B 48.0, and Protocol C 38.0, over test partitions of similar image count. Protocol C’s AUROC is therefore estimated on roughly a quarter as many independent patients as Protocol A’s, and part of its wider seed-to-seed spread reflects that rather than any property of the split rule. Subsampling participants does not correct for it, since AUROC is a rank statistic whose expectation barely moves with the number of clusters; the subject-level analysis above is the aggregation that does address it. The same asymmetry qualifies our recommendation of subject-level aggregation: averaging 5.35 predictions per patient under Protocol C removes substantially more variance than averaging 1.32 under Protocol A, so it does not neutralise the difference between protocols so much as redistribute it. Any single interval here should be read descriptively rather than as inference over a hypothetical larger seed population.
- **The secondary analyses support less than they appear to.** The identity probe uses one architecture and one probe family and reports a value that does not differ between leaky and honest protocols; it establishes that identity is decodable, across sessions as well as within them, not that it causes the gap. The sex-related disparity rests on five seeds and one cohort. The clinical translation is anchored to a single fixed-specificity operating point and to US-population prevalence, is computed on ADNI only, and is an arithmetic re-expression rather than a deployment estimate. The dose-response slope is specific to this cohort, this representation and the subject-substitution injection mechanism; the DenseNet-121 fit is noisier ($R^2 = 0.64$ against 0.85) and non-monotonic at $p = 1$, so we present it as a same-direction check rather than an independent confirmation, and we do not claim the slope generalises.

- **Disjointness is necessary but not sufficient, and our own gate demonstrates the gap.** The frozen ADNI splits this paper reports on pass every check the audit performs (zero subject, session and component overlap across partitions) while being severely stratified by component size, with training components spanning 4–6 scans and validation components 7–9 (§3.4.1). A split can therefore be provably leakage-free and still be a poor split, and the gate as specified cannot tell the difference. We regard this as the most useful negative result about the framework in the paper, because it was produced by the framework’s own output rather than argued in the abstract: a compositional check, comparing group-size, class and covariate distributions across partitions rather than only their intersection, belongs in the audit and was not in it. We have since added one. The released audit now reports the component-size distribution per partition and warns when the partitions are not drawn from comparable participants; run against the frozen manifests it fires immediately, flagging that validation components average 7.50 images against 4.65 in training and that the two ranges do not overlap. It is reported rather than blocking, deliberately: the manifests are frozen and every result here rests on them, so making the check a hard failure retroactively would invalidate the artefacts it is meant to describe. A caller generating new splits should treat the warning as a reason to regenerate. We did so as a control: regenerating the splits with components visited in shuffled rather than size-descending order removes the stratification (mean component size 5.15, 5.00 and 5.07 scans in train, validation and test) and gives a gap of +0.136 against +0.118. The same audit surfaces a second composition defect we did not fix, because the manifests are frozen: on the converter-inclusive arm the validation partition carries 20.7% AD images against 41.3% in training and 37.5% in test, in every seed. Thresholds for the operating-point and cost-of-leakage analyses are selected on that partition, so those two analyses inherit a validation sample whose class mix is not the test sample’s. Promoting the check to a blocking criterion is the first item of future work.
- **The audit gate certifies partition disjointness, nothing more.** A pass does not imply site- or scanner-independent generalisation: the site–label association we report (Cramér’s $V \approx 0.36$ in train rising to ≈ 0.54 in test) persists inside component-safe splits, and is a residual shortcut channel the audit exposes but does not remove. Harmonisation (ComBat-family methods [34, 35, 49], site-stratified splitting, leave-site-out evaluation) is complementary and outside this manuscript. Four of the audit’s advertised criteria are reported rather than enforced in this release (Table 4). The implementation covers structural MRI only, and its near-duplicate comparison is $\mathcal{O}(N^2)$, which is tractable at the scale studied here but would need locality-sensitive indexing beyond 10^5 images.

5 Conclusions

We treated evaluation validity in AD MRI deep learning as a question about data provenance and measured both links of that dependency. Random image-level evaluation overstates ADNI1 AUROC by +0.118 on the primary arm (paired-seed interval

[+0.096, +0.145] at $n = 5$, descriptive), and the gap survives every pre-specified variation of cohort, backbone, converter rule, acquisition type and input dimensionality (Figure 7; the volumetric arm in §3.3, where the gap is +0.235, about twice the 2D value).

Two controlled interventions locate the mechanism. Injecting contaminated test scans while holding the training set fixed gives +0.125 AUROC per unit overlap. Deleting subject identifiers at controlled rates shows how much leakage each protocol admits as provenance decays, with 49% of it prevented by the leakage graph at 10% identifier loss and subject-wise splitting returning 36% of test participants to training at that same level. A widely used public benchmark shows the same failure without any intervention: it is OASIS-1 redistributed under an identifier that merges and splits participants, and a split that respects that identifier still leaves +0.027 AUROC of inflation that nothing inside the release can reveal. A linear probe confirms that patient identity is decodable far above chance, across sessions as well as within them, but that quantity is unchanged between leaky and honest protocols and therefore establishes decodability rather than causation; the causal weight rests on the two interventions.

The field-level implication is that AD MRI benchmark results should be read as claims about an evaluation protocol, not only about a model architecture. The principal limitation is single-cohort headline scope: ADNI is the only cohort on which the cost-of-leakage, subgroup and dose-response analyses run, and external-cohort replication (AIBL, UK Biobank) is the immediate next step. SPLITGUARD-AD provides the reusable infrastructure to run it: frozen split manifests, an executable pre-training GO/NO-GO audit following Algorithm 1, and a benchmark report card.

Acknowledgements

Every training run reported here was executed in one code state, with the software environment of each node recorded in the released run logs (approximately 14.2 GPU-hours in total). Most arms ran on a single cloud-hosted NVIDIA H100 instance; the 50-run dose-response matrix was rerun in that same code state on an NVIDIA RTX PRO 4500 instance and the 15-run volumetric arm on an NVIDIA RTX 2000 Ada instance, so those arms and the ones they are compared with differ in hardware but not in code. We report the slope as a descriptive calibration constant and not as a transferable coefficient, which is how it is read throughout. Data used in preparation of this article were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (<https://adni.loni.usc.edu>). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data but did not participate in the analysis or writing of this report. A complete listing of ADNI investigators can be found at https://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf. OASIS-1 data were provided in part by OASIS Cross-Sectional Principal Investigators: D. Marcus, R. Buckner, J. Csernansky, J. Morris (P50 AG05681, P01 AG03991, P01 AG026276, R01 AG021910, P20 MH071616, U24 RR021382).

Declarations

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Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval and consent to participate

This study used retrospective, de-identified secondary imaging data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) and the Open Access Series of Imaging Studies (OASIS-1), including a public Kaggle benchmark that we show to be a redistribution of OASIS-1 images. No new human participants were recruited, no new imaging was acquired, and no direct participant contact occurred. ADNI and OASIS obtained ethics approval and informed consent at the original data-collection sites; the present secondary analysis was conducted under their respective data-use agreements and public-access terms. The Kaggle benchmark consists of already-public OASIS-1 slice images, and its provenance recovery was performed against OASIS-1 data under the same terms; the recovered image-to-participant mapping is not redistributed. Because this analysis is limited to secondary de-identified data, institutional review board (IRB) approval was not separately sought at Istanbul Medipol University in accordance with local policy on secondary de-identified data reuse. Participant-level manifests remain local and are hashed via PBKDF2 with a per-release salt so as not to support cross-dataset participant re-identification; only aggregate result tables are redistributed.

Consent for publication

No individual participant data, images, or identifying details are reproduced in this manuscript or in any released artefact, so participant consent for publication is not applicable. All released tables are aggregate, and participant identifiers in the Tier-3 manifests are replaced by a PBKDF2 derivation under a secret salt.

Use of generative AI

Large-language-model coding assistants were used to implement and revise the released software and to draft manuscript prose; §2.4 documents this in the Methods, as the journal requires, and states what the tools did and did not do. The study design, the protocol definitions and the interpretation of every result are the authors’; no reported number was produced or altered by an LLM. The authors reviewed and edited all such output and take full responsibility for the content of this publication.

Author contributions

Mehmet Berke Isler: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Irem**

Iltter: Investigation (OASIS-1 cohort), Formal analysis, Writing – review & editing.
Mehmet Kemal Ozdemir: Supervision, Methodology, Writing – review & editing,
Project administration.

Data and code availability

The SPLITGUARD-AD framework, the training and analysis pipeline, and the manuscript sources are released at <https://github.com/mehmetberkeisler/splitguard-ad> under the Apache-2.0 licence.

The release boundary is aggregate-only: scripts, paper sources, figures, aggregate result tables and rebuild instructions are public, while raw data, model weights, local manifests, detailed audit rows and per-image predictions remain local. The manifest release boundary is tier-dependent, and the artefacts for each tier are materialised by `scripts/build_release_artefacts.py` into the tracked `release/` directory, with a `RELEASE_MANIFEST.json` recording the row counts and the columns withheld per tier. **Tier 1 (redistributed public benchmark):** the release includes the frozen filename-key split manifest at image-path level (6,400 rows), its leakage-graph component table (6,400 rows) and the near-duplicate candidate list (1,582 pairs), which together document the split whose contamination §2.3.1 measures. The participant-level mapping to OASIS-1 is not released; `scripts/audit_tier1_ground_truth.py` and `scripts/tier1_split_rules_vs_truth.py` regenerate it, the recovered-participant splits and every Tier-1 contamination and calibration number from a user’s own Kaggle and OASIS-1 downloads. **Tier 2 (OASIS-1):** image-level identifiers are not ours to redistribute but OASIS-1 access is open to any registered user, so the release includes the frozen split manifest (1,175 rows) and component table (2,180 rows) indexed by the public OASIS subject, session and volume codes only, with no filesystem paths and no clinical fields (CDR, MMSE, education, SES, age, sex), which a third party re-derives from their own download. The projection is enforced in code rather than by convention: the release builder refuses to emit any column on its per-tier restricted list. **Tier 3 (ADNI1):** under the LONI/ADNI data-use agreement, participant-level identifiers cannot be redistributed. For this tier the release includes *hashed* component-safe manifests, one per frozen split of the five-seed primary arm (220 rows each, one row per single-subject component): each row carries a PBKDF2-HMAC-SHA256 derivation of the subject identifier (600,000 iterations under a secret per-release salt), an integer component-size summary, the binary CN/AD label, and the non-identifying acquisition fields (`scanner_field_strength`, modality flag), but no participant-level identifiers, dates, or image paths. The converter-inclusive arm is not released in this form, because a converting participant contributes both CN and AD scans to one component and the aggregation refuses a component whose label is not uniform rather than choosing one for it. Key stretching rather than a plain digest is required here: ADNI PTIDs are drawn from a space of roughly 10^7 candidates, so a single-round hash of one is invertible by enumeration in under a second, and the salt cannot be published for the same reason. A downstream auditor

can therefore verify component sizes, class balance, and cross-partition contamination on Tier 3 without obtaining the underlying ADNI data, and without the release itself re-identifying participants. Data access for the three tiers: Tier 1 is the public Kaggle four-class Alzheimer’s MRI dataset [37], itself derived from OASIS-1; Tier 2 (OASIS-1) [38] is available at <https://www.oasis-brains.org/>; Tier 3 (ADNI1: Complete 3Yr 1.5T) is accessed through the LONI Image and Data Archive (IDA) at <https://ida.loni.usc.edu/> under an approved Data Use Agreement. ADNI participant-level study files and image volumes are retained locally and are not redistributed; only the hashed manifests, aggregate result tables, and rebuild scripts are released for that tier.

Reproducibility boundary of the reported runs

Every AUROC in this manuscript comes from `scripts/gpu_program.py` in a single code state, on the two GPUs named in the Acknowledgements, with seeded samplers, per-worker augmentation seeding and deterministic cuDNN kernels; the environment (driver, CUDA, cuDNN and every package version) is recorded with the run. Each run persists its per-image test predictions. `scripts/gpu_postprocess.py` derives the headline AUROCs, intervals, controls and result tables from those predictions and writes them into generated files the manuscript reads, so those numbers are never transcribed by hand; the remaining quantities are computed by the analysis scripts and held to their artefacts by `scripts/verify_paper_numbers.py`, which fails on any divergence. Those per-image predictions carry ADNI participant identifiers and therefore stay local; what is released is the code that produces them, the aggregate outputs, and `scripts/verify_paper_numbers.py`, which re-derives every numerical claim from those outputs and fails the build on any divergence. Model weights are not released, because weights trained on participant-level data fall under the same data-use terms as the data. Re-running training on different hardware reproduces the protocol ordering, the sign and the approximate magnitude of every effect, but not the third decimal place, and we can state that from measurement rather than expectation: regenerating every arm moved 36 of the 116 compared runs by at least 0.010 AUROC and one of them by 0.084, while every protocol ordering, every gap sign and every interval-level conclusion held. The run-by-run comparison is released as `reports/rerun_comparison.md`, and because each downstream artefact is regenerated from the predictions rather than edited, the manuscript follows those movements automatically.

Supplementary Material

Supplementary Material accompanying this submission contains: (S1) consolidated training hyperparameters for all tiers; (S2) the ten-row ADNI operating-point sensitivity table; (S3) SPLITGUARD-AD outputs mapped to the CLAIM [24], TRI-POD+AI [26] and PROBAST domain 4 [27] reporting domains; (S4) the dose-response mixed-effects model specification and cluster-robust inference; (S5) the ADNI 180-day linkage-integrity artefact schema; and (S6) the sex-stratified ADNI analysis.

References

- [1] Korolev S, Safiullin A, Belyaev M, Dodonova Y. Residual and plain convolutional neural networks for 3D brain MRI classification. In: 2017 IEEE 14th International Symposium on Biomedical Imaging (ISBI); 2017. p. 835–838.
- [2] Lian C, Liu M, Zhang J, Shen D. Hierarchical Fully Convolutional Network for Joint Atrophy Localization and Alzheimer’s Disease Diagnosis Using Structural MRI. *IEEE Transactions on Pattern Analysis and Machine Intelligence*. 2020;42(4):880–893. <https://doi.org/10.1109/TPAMI.2018.2889096>.
- [3] Qiu S, Joshi PS, Miller MI, Xue C, Zhou X, Karjadi C, et al. Development and validation of an interpretable deep learning framework for Alzheimer’s disease classification. *Brain*. 2020;143(6):1920–1933. <https://doi.org/10.1093/brain/awaa137>.
- [4] World Health Organization. Dementia. WHO; 2023. Available from: <https://www.who.int/news-room/fact-sheets/detail/dementia>.
- [5] Jack CR, Albert MS, Knopman DS, McKhann GM, Sperling RA, Carrillo MC, et al. Introduction to the recommendations from the National Institute on Aging-Alzheimer’s Association workgroups on diagnostic guidelines for Alzheimer’s disease. *Alzheimer’s & Dementia*. 2011;7(3):257–262. <https://doi.org/10.1016/j.jalz.2011.03.004>.
- [6] Young VM, Gates S, Garcia LY, Salardini A. Data Leakage in Deep Learning for Alzheimer’s Disease Diagnosis: A Scoping Review of Methodological Rigor and Performance Inflation. *Diagnostics*. 2025;15(18):2348. <https://doi.org/10.3390/diagnostics15182348>.
- [7] Wen J, Thibeu-Sutre E, Diaz-Melo M, Samper-González J, Routier A, Bottani S, et al. Convolutional neural networks for classification of Alzheimer’s disease: Overview and reproducible evaluation. *Medical Image Analysis*. 2020;63:101694. <https://doi.org/10.1016/j.media.2020.101694>.
- [8] Yagis E, García Seco de Herrera A, Citi L. Effect of data leakage in brain MRI classification using 2D convolutional neural networks. *Scientific Reports*. 2021;11(1):22544. <https://doi.org/10.1038/s41598-021-01681-w>.
- [9] Rosenblatt M, Tejavibulya L, Jiang R, Noble S, Scheinost D. Data leakage inflates prediction performance in connectome-based machine learning models. *Nature Communications*. 2024;15(1):1829. <https://doi.org/10.1038/s41467-024-46150-w>.
- [10] Geirhos R, Jacobsen JH, Michaelis C, Zemel R, Brendel W, Bethge M, et al. Shortcut learning in deep neural networks. *Nature Machine Intelligence*. 2020;2:665–673. <https://doi.org/10.1038/s42256-020-00257-z>.
- [11] Zech JR, Badgeley MA, Liu M, Costa AB, Titano JJ, Oermann EK. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*. 2018;15(11):e1002683. <https://doi.org/10.1371/journal.pmed.1002683>.
- [12] DeGrave AJ, Janizek JD, Lee SI. AI for radiographic COVID-19 detection selects shortcuts over signal. *Nature Machine Intelligence*. 2021;3:610–619. <https://doi.org/10.1038/s42256-021-00338-7>.

- [13] Oakden-Rayner L, Dunnmon J, Carneiro G, Ré C. Hidden Stratification Causes Clinically Meaningful Failures in Machine Learning for Medical Imaging. In: Proceedings of the ACM Conference on Health, Inference, and Learning (CHIL); 2020. p. 151–159.
- [14] Schwarz CG, Kremers WK, Therneau TM, Sharp RR, Gunter JL, Vemuri P, et al. Identification of Anonymous MRI Research Participants with Face-Recognition Software. *New England Journal of Medicine*. 2019;381(17):1684–1686. <https://doi.org/10.1056/NEJMc1908881>.
- [15] Wachinger C, Golland P, Kremen W, Fischl B, Reuter M. BrainPrint: A discriminative characterization of brain morphology. *NeuroImage*. 2015;109:232–248. <https://doi.org/10.1016/j.neuroimage.2015.01.032>.
- [16] Puglisi L, Eshaghi A, Parker GJM, Barkhof F, Alexander DC, Ravi D. Deep-BrainPrint: A novel contrastive framework for brain MRI re-identification. *Proceedings of Machine Learning Research, Medical Imaging with Deep Learning*. 2023;.
- [17] Liem F, Varoquaux G, Kynast J, Beyer F, Kharabian Masouleh S, Huntenburg JM, et al. Predicting brain-age from multimodal imaging data captures cognitive impairment. *NeuroImage*. 2017;148:179–188. <https://doi.org/10.1016/j.neuroimage.2016.11.005>.
- [18] Roberts M, Driggs D, Thorpe M, Gilbey J, Yeung M, Ursprung S, et al. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nature Machine Intelligence*. 2021;3:199–217. <https://doi.org/10.1038/s42256-021-00307-0>.
- [19] Varoquaux G, Cheplygina V. Machine learning for medical imaging: methodological failures and recommendations for the future. *npj Digital Medicine*. 2022;5:48. <https://doi.org/10.1038/s41746-022-00592-y>.
- [20] Kapoor S, Narayanan A. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*. 2023;4(9):100804. <https://doi.org/10.1016/j.patter.2023.100804>.
- [21] Pedregosa F, Varoquaux G, Gramfort A, Michel V, Thirion B, Grisel O, et al. Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*. 2011;12:2825–2830.
- [22] Cardoso MJ, et al.: MONAI: An open-source framework for deep learning in healthcare. *arXiv preprint arXiv:2211.02701*.
- [23] Joeres R, Blumenthal DB, Kalinina OV. Data splitting to avoid information leakage with DataSAIL. *Nature Communications*. 2025;16(3337). <https://doi.org/10.1038/s41467-025-58606-8>.
- [24] Tejani AS, Klontzas ME, Gatti AA, Mongan JT, Moy L, Park SH, et al. Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 Update. *Radiology: Artificial Intelligence*. 2024;6(4):e240300. <https://doi.org/10.1148/ryai.240300>.
- [25] Mongan J, Moy L, Kahn CE Jr. Checklist for Artificial Intelligence in Medical Imaging (CLAIM): A Guide for Authors and Reviewers. *Radiology: Artificial Intelligence*. 2020;2(2):e200029. <https://doi.org/10.1148/ryai.2020200029>.

- [26] Collins GS, Moons KGM, Dhiman P, Riley RD, Beam AL, Van Calster B, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning. *BMJ*. 2024;385:e078378. <https://doi.org/10.1136/bmj-2023-078378>.
- [27] Wolff RF, Moons KGM, Riley RD, Whiting PF, Westwood M, Collins GS, et al. PROBAST: A Tool to Assess the Risk of Bias and Applicability of Prediction Model Studies. *Annals of Internal Medicine*. 2019;170(1):51–58. <https://doi.org/10.7326/M18-1376>.
- [28] Rumala DJ. How You Split Matters: Data Leakage and Subject Characteristics Studies in Longitudinal Brain MRI Analysis. In: *Clinical Image-Based Procedures, Fairness of AI in Medical Imaging, and Ethical and Philosophical Issues in Medical Imaging (FAIMI 2023)*. vol. 14242 of *Lecture Notes in Computer Science*. Springer Nature Switzerland; 2023. p. 235–245.
- [29] Ghiasi MM, Falck RS, Liu-Ambrose T, Tam RC. Explainable machine learning for predicting longitudinal dementia status: Establishing a leakage-free benchmark. *PLOS Digital Health*. 2026;5(5):e0001409. <https://doi.org/10.1371/journal.pdig.0001409>.
- [30] Ellis KA, Bush AI, Darby D, De Fazio D, Foster J, Hudson P, et al. The Australian Imaging, Biomarkers and Lifestyle (AIBL) study of aging: methodology and baseline characteristics of 1112 individuals recruited for a longitudinal study of Alzheimer’s disease. *International Psychogeriatrics*. 2009;21(4):672–687. <https://doi.org/10.1017/S1041610209009405>.
- [31] Sudlow C, Gallacher J, Allen N, Beral V, Burton P, Danesh J, et al. UK Biobank: An Open Access Resource for Identifying the Causes of a Wide Range of Complex Diseases of Middle and Old Age. *PLOS Medicine*. 2015;12(3):e1001779. <https://doi.org/10.1371/journal.pmed.1001779>.
- [32] LaMontagne PJ, Benzinger TLS, Morris JC, Keefe S, Hornbeck R, Xiong C, et al. OASIS-3: Longitudinal Neuroimaging, Clinical, and Cognitive Dataset for Normal Aging and Alzheimer Disease. *medRxiv*. 2019;<https://doi.org/10.1101/2019.12.13.19014902>.
- [33] Glocker B, Robinson R, Castro DC, Dou Q, Konukoglu E.: Machine Learning with Multi-Site Imaging Data: An Empirical Study on the Impact of Scanner Effects. *Medical Imaging meets NeurIPS Workshop*; arXiv:1910.04597.
- [34] Johnson WE, Li C, Rabinovic A. Adjusting batch effects in microarray expression data using empirical Bayes methods. *Biostatistics*. 2007;8(1):118–127. <https://doi.org/10.1093/biostatistics/kxj037>.
- [35] Fortin JP, Parker D, Tung B, Watanabe T, Elliott MA, Ruparel K, et al. Harmonization of multi-site diffusion tensor imaging data. *NeuroImage*. 2017;161:149–170. <https://doi.org/10.1016/j.neuroimage.2017.08.047>.
- [36] Pomponio R, Erus G, Habes M, Doshi J, Srinivasan D, Mamourian E, et al. Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan. *NeuroImage*. 2020;208:116450. <https://doi.org/10.1016/j.neuroimage.2019.116450>.

- [37] : Alzheimer’s Dataset (4 class of images). 6,400 2D brain-MRI JPEG images in four severity folders, uniform 176×208 pixels, released without subject identifiers, acquisition metadata or a statement of origin. Shown in this work by content matching to be 32 axial slices each of 200 OASIS-1 participants, with folder labels equal to their CDR. Kaggle public dataset. Available from: <https://www.kaggle.com/datasets/tourist55/alzheimers-dataset-4-class-of-images>.
- [38] Marcus DS, Wang TH, Parker J, Csernansky JG, Morris JC, Buckner RL. Open Access Series of Imaging Studies (OASIS): Cross-sectional MRI Data in Young, Middle Aged, Nondemented, and Demented Older Adults. *Journal of Cognitive Neuroscience*. 2007;19(9):1498–1507. <https://doi.org/10.1162/jocn.2007.19.9.1498>.
- [39] Weiner MW, Veitch DP, Aisen PS, Beckett LA, Cairns NJ, Green RC, et al. The Alzheimer’s Disease Neuroimaging Initiative: A review of papers published since its inception. *Alzheimer’s & Dementia*. 2012;8(1 Suppl):S1–S68. <https://doi.org/10.1016/j.jalz.2011.09.172>.
- [40] Jack CR, Bernstein MA, Fox NC, Thompson P, Alexander G, Harvey D, et al. The Alzheimer’s Disease Neuroimaging Initiative (ADNI): MRI methods. *Journal of Magnetic Resonance Imaging*. 2008;27(4):685–691. <https://doi.org/10.1002/jmri.21049>.
- [41] Petersen RC, Aisen PS, Beckett LA, Donohue MC, Gamst AC, Harvey DJ, et al. Alzheimer’s Disease Neuroimaging Initiative (ADNI): Clinical characterization. *Neurology*. 2010;74(3):201–209. <https://doi.org/10.1212/WNL.0b013e3181cb3e25>.
- [42] Efron B, Tibshirani R. Bootstrap methods for standard errors, confidence intervals, and other measures of statistical accuracy. *Statistical Science*. 1986;1(1):54–75. <https://doi.org/10.1214/ss/1177013815>.
- [43] Carpenter J, Bithell J. Bootstrap confidence intervals: when, which, what? A practical guide for medical statisticians. *Statistics in Medicine*. 2000;19(9):1141–1164. [https://doi.org/10.1002/\(SICI\)1097-0258\(20000515\)19:9<1141::AID-SIM479>3.0.CO;2-F](https://doi.org/10.1002/(SICI)1097-0258(20000515)19:9<1141::AID-SIM479>3.0.CO;2-F).
- [44] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*; 2016. p. 770–778.
- [45] Niculescu-Mizil A, Caruana R. Predicting good probabilities with supervised learning. In: *Proceedings of the 22nd International Conference on Machine Learning (ICML)*; 2005. p. 625–632.
- [46] Guo C, Pleiss G, Sun Y, Weinberger KQ. On calibration of modern neural networks. In: *Proceedings of the 34th International Conference on Machine Learning (ICML)*. vol. 70 of *Proceedings of Machine Learning Research*; 2017. p. 1321–1330. Available from: <https://proceedings.mlr.press/v70/guo17a.html>.
- [47] Van Calster B, McLernon DJ, van Smeden M, Wynants L, Steyerberg EW. Calibration: the Achilles heel of predictive analytics. *BMC Medicine*. 2019;17(1):230. <https://doi.org/10.1186/s12916-019-1466-7>.

- [48] Bergsma W. A bias-correction for Cramér’s V and Tschuprow’s T . Journal of the Korean Statistical Society. 2013;42(3):323–328. <https://doi.org/10.1016/j.jkss.2012.10.002>.
- [49] Bashyam VM, Doshi J, Erus G, Srinivasan D, Abdulkadir A, Singh A, et al. Deep Generative Medical Image Harmonization for Improving Cross-Site Generalization in Deep Learning Predictors. Journal of Magnetic Resonance Imaging. 2022;55(3):908–916. <https://doi.org/10.1002/jmri.27908>.
- [50] Rouzrokh P, Khosravi B, Faghani S, Moassefi M, Vera Vázquez D, Singh Y, et al. Mitigating Bias in Radiology Machine Learning: 1. Data Handling. Radiology: Artificial Intelligence. 2022;4(5):e210290. <https://doi.org/10.1148/ryai.210290>.
- [51] Hua X, Hibar DP, Lee S, Toga AW, Jack CR Jr, Weiner MW, et al. Sex and age differences in atrophic rates: an ADNI study with $n = 1368$ MRI scans. Neurobiology of Aging. 2010;31(8):1463–1480. <https://doi.org/10.1016/j.neurobiolaging.2010.04.033>.
- [52] Sundermann EE, Maki PM, Rubin LH, Lipton RB, Landau S, Biegon A. Female advantage in verbal memory: Evidence of sex-specific cognitive reserve. Neurology. 2016;87(18):1916–1924. <https://doi.org/10.1212/WNL.0000000000003288>.
- [53] Mielke MM, Vemuri P, Rocca WA. Clinical epidemiology of Alzheimer’s disease: assessing sex and gender differences. Clinical Epidemiology. 2014;6:37–48. <https://doi.org/10.2147/CLEP.S37929>.
- [54] Rajan KB, Weuve J, Barnes LL, McAninch EA, Wilson RS, Evans DA. Population estimate of people with clinical Alzheimer’s disease and mild cognitive impairment in the United States (2020–2060). Alzheimer’s & Dementia. 2021;17(12):1966–1975. <https://doi.org/10.1002/alz.12362>.
- [55] Thomas G, Smith EE, Ismail Z, Kirkham JG, Roach P, Goodarzi Z, et al. Potential Eligibility for Disease Modifying Therapies Among Individuals Referred to A Tertiary Care Memory Clinic in Calgary, Alberta, Canada: Implications for AD Biomarker Testing (Prospective Registry for Persons with Memory Symptoms, PROMPT). Alzheimer’s & Dementia. 2024;<https://doi.org/10.1002/alz.091118>.
- [56] Jack CR, Bennett DA, Blennow K, Carrillo MC, Dunn B, Haeberlein SB, et al. NIA-AA Research Framework: Toward a biological definition of Alzheimer’s disease. Alzheimer’s & Dementia. 2018;14(4):535–562. <https://doi.org/10.1016/j.jalz.2018.02.018>.
- [57] Lin JS, O’Connor E, Rossom RC, Perdue LA, Eckstrom E. Screening for cognitive impairment in older adults: A systematic review for the U.S. Preventive Services Task Force. Annals of Internal Medicine. 2013;159(9):601–612. <https://doi.org/10.7326/0003-4819-159-9-201311050-00730>.
- [58] Wittenberg R, Knapp M, Hu B, Comas-Herrera A, King D, Rehill A, et al. The costs of dementia in England. International Journal of Geriatric Psychiatry. 2019;34(7):1095–1103. <https://doi.org/10.1002/gps.5113>.